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# Customer concentration and stock price crash risk

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ABSTRACT

We investigate the impact of customer concentration on stock price crash risk. Customer concentration may represent a source of significant cash flow and business risk for supplier firms or benefit supplier firms in terms of efficient product, inventory and supply chain management. Using a large sample of firms for the period 1977–2016, we find that corporate customer concentration is positively associated with stock price crash risk. We find, however, that government customer concentration is negatively associated with stock price crash risk. Further analyses demonstrate that our results are robust to potential endogeneity and reverse causality concerns. In the cross-sectional analyses, we find that the likelihood that a supplier firm with a concentrated customer base experiences higher crash risk is attenuated by lower switching cost and is accentuated when the degree of information asymmetry is high.

## 1. Introduction

An important risk factor that influences a firm's ability to generate current and future cash flow is the concentration of its customer base. According to the Statement of Financial Accounting Standards (SFAS) No. 131, firms are required to disclose information about their customers[1] who individually represent more than 10% of the firm's revenues (hereinafter "major customers"), reflecting the view that customer concentration is a significant source of a firm's operational risk. On the one hand, a supplier firm[2] with higher customer concentration would face relatively high volatility of current and future cash flows and other business risks (Dhaliwal, Judd, Serfling, & Shaikh, 2016; Dhaliwal, Michas, Naiker, & Sharma, 2019; Huang, Lobo, Wang, & Xie, 2016). This is because supplier cash flows rely excessively on a few concentrated customers, which makes it difficult for the supplier to diversify the associated operational risk. On the other hand, such supplier firms would be better able to reduce inventory cost and improve the efficiency in working capital management. This is because customer concentration allows for more stable customer-supplier relationships and thus helps to reduce the information asymmetry that may exist along the supply chain (Kalwani & Narayandas, 1995; Patatoukas, 2012).

In this study, we investigate the association between customer concentration and stock price crash risk of a supplier firm. Specifically, we first examine whether corporate customer concentration of a supplier firm is associated with its future stock price crash risk. Second, we further examine how this customer concentration-stock price crash risk association is different when the customers are affiliated with the government.

Stock price crash risk refers to the likelihood that an abrupt, large-scale decline in a firm's stock price could occur, and is typically characterized by the third moment of firm-specific return distribution or the conditional skewness of distribution of firm-specific returns after netting out common (market-wide and/or industry-wide) factors. The prominent predictor of stock price crash risk is the managerial tendency to withhold bad news from outside investors. However, managers can hide and accumulate bad news within the firm only up to a certain threshold. When the hidden bad news accumulated over time reaches a tipping point beyond which the cost of hiding that same bad news exceeds the associated benefit, managers are forced to exercise an abandonment option, which results in all the hidden, accumulated bad news to come out all at once. As a result, we observe an abrupt, large-scale decline in firm-specific return or a stock price crash (Jin & Myers, 2006).

On the one hand, corporate customer concentration can increase the likelihood of stock price crash occurrence or simply, stock price crash

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[1] Major customers can refer to either major corporate customers or major government customers throughout the paper, depending on the type of customers.
[2] We use 'a supplier firm' and 'a supplier' interchangeably throughout the paper.

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risk. First, a supplier firm with a concentrated corporate customer base, who relies typically on a few major customers, faces significant business and cash flow risks, if any major customer becomes distressed or switches to a different supplier, or internally develops products (Dhaliwal et al., 2016, 2019; Huang et al., 2016). Further, a supplier firm with a concentrated corporate customer base has a greater tendency to engage in relationship-specific investments (Banker, Byzalov, & Plehn-Dujowich, 2014; Kang, Mahoney, & Tan, 2009). Such investments are, in general, risky, particularly when the supplier-customer relationship becomes unstable or adversely affected by uncontrollable external factors. In addition, the relationship-specific investments are more likely to involve negative net present value (NPV) projects that entail short-term losses and are difficult to abandon (Irvine, Park, & Yildizhan, 2016). A supplier with a concentrated corporate customer base has greater business and cash flow risks, and in this vein, is likely to experience a high level of bad news hoarding, leading to stock price crash risk in the future. Second, such a supplier firm may have incentive to manage earnings upward to maintain its business relationships with major customers having a bargaining advantage (Raman & Shahrur, 2008). Furthermore, the supplier may have incentive to manage these major customers' perceptions, particularly adverse perceptions, of its ability to generate future cash flows and manage business risk (Dhaliwal et al., 2016; Huang et al., 2016). As a result, managers of such supplier firms are more likely to engage in earnings management and bad news hoarding to manage major customers' expectations, which in turn increases its future stock price crash risk.

On the other hand, corporate customer concentration can decrease stock price crash risk. First, a supplier firm's reliance on a few major customers may lead to a more efficient supplier-customer relationship and thus improve its financial performance (Irvine et al., 2016; Patatoukas, 2012). This decreases the business and cash flow risks faced by the supplier firm and bad news hoarding, which in turn contributes to lowering its future stock price crash risk. Second, a supplier firm with major corporate customers should be more responsive to the demand for higher quality accounting practices and conservative accounting practices that facilitate external monitoring of the supplier firm by those major customers (Hui, Klasa, & Yeung, 2012). Given that accounting conservatism reduces stock price crash risk (Kim & Zhang, 2016), as it deters bad news hoarding and overinvestment, a supplier firm with a concentrated corporate customer base is likely to experience a lower level of crash risk.

In contrast to corporate customer concentration at a general level, we argue that suppliers' reliance on government or government-related customers decreases stock price crash risk. This is because business with government customers, which is typically characterized by long-term contractual relationships and cost-plus pricing, has a very low downside risk such as bankruptcy risk, and can thus provide suppliers with stable cash flows (Irvine et al., 2016; Patatoukas, 2012). Thus, the stable cash flow and reduced business risk related to business with a concentrated government customer base contribute to lowering the bad news hoarding and the stock price crash risk of the supplier firm.

Following the Statement of Financial Accounting Standards No. 14 (SFAS 14) and prior studies (Dhaliwal et al., 2016; Patatoukas, 2012), we construct three measures to capture the degree of a supplier firm's customer concentration: (1) an indicator variable which has a value of 1 if a supplier firm reports at least one customer who accounts for more than 10% of its annual revenues; (2) the sales-based Herfindahl-Hirschman Index as the sum of the squares of the ratios of supplier firm revenues from major customer to its total revenues; and (3) the fraction of a supplier firm's annual revenues that are accounted for by all major customers. For firm-specific measures of stock price crash risk, we also follow prior research (e.g., Hutton, Marcus, & Tehranian, 2009; Kim et al., 2011a, 2011b): (1) an indicator variable which has a value of 1 if we observe at least one crash week during a year; (2) the negative of the ratio of the third moment to the standard deviation cubed of firm-specific abnormal weekly returns; and (3) the Log Ratio of the standard deviation of firm-specific abnormal weekly returns of the down weeks, compared to those of the up weeks.

Our findings based on the sample of 131,505 firm-year observations (13,612 supplier firms) for the years 1977–2016 are as follows. First, we find that the corporate customer concentration of a supplier firm is positively associated with its future stock price crash risk, and that this association is robust to the use of alternative measures of corporate customer concentration and stock price crash risk. Second, we find that government customer concentration is negatively associated with stock price crash risk and that this association is also robust to the use of alternative measures of government customer concentration and stock price crash risk. Our findings suggest that a supplier firm with a highly concentrated corporate customer base is more vulnerable to business and cash flow risks and that its managers are therefore likely incentivised to engage in earnings management to manage corporate customers' perceptions, which leads to increased bad news hoarding and subsequent heightened stock price crash risk. However, a supplier firm with more stable and less risky government customers is less vulnerable to business and cash flow risks and its managers are therefore less likely to hoard bad news, which lowers their stock price crash risks.

In addition, we examine cross-sectional differences in the association between corporate customer concentration and stock price crash risk. First, when major corporate customers can easily switch supplier firms at a low cost, major corporate customers can act as an additional external monitoring mechanism and managers of the supplier firm which relies on a few major corporate customers are more likely to disclose information in a timely manner and are less able to hoard bad news. Alternatively, low switching costs make it easier for major customers to switch their suppliers and thereby increase the business and cash flow risks of the suppliers and their subsequent bad news hoarding, resulting in increased stock price crash risks. To test this notion, we construct an interaction variable between corporate customer concentration and low switching cost, which is an indicator variable that equals 1 if a supplier firm's market share is below the median market share of all firms excluding the supplier firm in the same 2-digit SIC industry in a given year, and 0 otherwise. We then conduct regression analyses and find that the likelihood of a supplier firm with a concentrated corporate customer base having a higher stock price crash risk is attenuated when major corporate customers face lower costs of switching suppliers, suggesting that a supplier firm with a concentrated corporate customer base is less likely to hoard unfavourable news when major corporate customers have a lower switching cost, leading to lower stock price crash risks overall.

Second, the detrimental effect of corporate customer concentration of a supplier firm on its future stock price crash risk may be more pronounced when the supplier firm is subject to greater information asymmetry. To test this notion, we construct an interaction variable between corporate customer concentration and low information asymmetry. We measure the information asymmetry using analyst forecast dispersion. Specifically, we create an indicator variable that equals 1 if the dispersion of annual EPS forecasts by financial analysts following a supplier firm is below the 2-digit SIC industry median dispersion in a given year, and 0 otherwise. We find that the detrimental impact of corporate customer concentration on future stock crash risk is more pronounced for supplier firms with higher information asymmetry.

We conduct propensity score matching analyses to estimate causal treatment effects and the results using the PSM subsample corroborate the conclusion that corporate customer concentration leads supplier firms to experience greater stock price crash. It is possible that our main results might be driven by certain unobservable firm characteristics which are omitted in the models. To mitigate the effect of the omitted-variable bias, we estimate the firm fixed-effect model in place of the industry fixed effects. Results obtained with the firm fixed-effect model are mostly consistent with our main results. Thus, our results are not

likely to be driven by unobservable and time-invariant firm characteristics which can be omitted. Furthermore, in order to mitigate endogeneity that can be attributed to reverse causality and unobserved heterogeneity, we conduct an instrumental-variable (IV) analysis. Following prior studies (Hilary & Hui, 2009; Kumar, Page, & Spalt, 2011) that use an earlier year's value of their focal explanatory variable as an IV, we employ customer concentration in the earliest year for each firm as our IV assuming that it affects customer concentration in any subsequent years of the same firm (relevance condition), but could influence the firm's stock crash risk in subsequent years only through its impact on subsequent years' customer concentration (exclusion condition). As an additional IV, we utilise the average corporate customer concentration in the same U.S. county, excluding firm *i* because firms in the same area should attract similar customers and exhibit similar customer concentration (relevance condition), but do not affect the future stock crash risk (exclusion condition). Our IV analysis confirms that greater corporate customer concentration leads to higher stock price crash risk. It can be claimed that our results could pick up an indirect effect of corporate customer concentration via corporate tax avoidance on stock crash risk. To address this issue, we establish the mediation models where we employ OLS regressions where two of the stock crash risk measures (DUVOL and NCSKEW) are regressed on each corporate customer concentration measure while corporate tax avoidance behaviour is included as a mediating variable. We find that corporate customer concentration is a direct determinant of stock crash risk, rather than a mere reflection of its indirect effect via corporate tax avoidance.

Our paper differs from the concurrent paper (Chen, Hu, Yao, & Zhao, 2018) in several aspects. First, we hypothesize a negative relation between government customers and stock price crash risk and find that the suppliers' reliance on government or government-related customers decreases stock price crash risk due to the low downside risk of governmental customers. Second, we develop hypotheses based on business risk and accounting quality in a more organized manner. This differs from Chen, Hu, Yao, and Zhao (2018) who developed their hypothesis based only on earnings quality. Our study develops hypotheses using both earnings quality and business risk. That is, we clearly and separately frame our argument around how the business risk associated with customer concentration affects stock crash risk and how the earnings quality incentives induced by customer concentration affect stock price crash risk.

Third, our cross-sectional analyses investigate how the detrimental relationship between corporate customer concentration and stock price crash risk differs when major corporate customers can easily switch their supplier firms at a low cost and that the supplier firm is subject to greater information asymmetry whereas Chen et al.(2018) analyse the cross-sectional effects on durable goods and high R&D expenses. Fourth, following Chen, Hong, and Stein (2001), we use the down-to-up volatility (*DUVOL*) as an additional stock crash risk measure, given that, unlike other stock crash risk measures, *DUVOL* is less likely to be overly influenced by a handful of extreme weekly returns because it measures return asymmetries that do not involve third moments of weekly returns. Lastly, we demonstrate that corporate customer concentration acts as a direct determinant of stock price crash risk, rather than as a mere reflection of its indirect effect via corporate tax avoidance.

This study contributes to the current body of literature in several ways. Recent studies investigate how customer concentration affects suppliers' financial policy (Banerjee, Dasgupta, & Kim, 2008; Itzkowitz, 2013), performance and risk (Campello & Gao, 2017; Dhaliwal et al., 2016; Gosman, Kelly, Olsson, & Warfield, 2004; Kalwani & Narayandas, 1995; Patatoukas, 2012), financial reporting quality (Hui et al., 2012; Raman & Shahrur, 2008), and auditor decision making (Dhaliwal et al., 2019; Krishnan, Patatoukas, & Wang, 2019). Particularly, some studies examine the association between customer concentration and stock market performance and risk. Patatoukas (2012) and Irvine et al. (2016) show a positive association between changes in customer concentration and suppliers' stock returns and suggest that changes in customer concentration are incrementally informative to investors, when setting stock prices.[3] In contrast, Dhaliwal et al. (2016) document that suppliers with a concentrated customer base have a higher cost of equity. To our knowledge, however, prior research has paid little attention to the effect of customer concentration on future stock price crash risk. As a result, little is known about whether and how suppliers' crash risk is influenced by the extent of their corporate customer concentration. Unlike the above two studies which focus on the first moment of return distribution, i.e., stock return performance and cost of equity, our study focuses primarily on the effect of customer concentration on stock price crash risk, which is the conditional skewness or extreme *negative tail risk* of stock return distribution (third moment). As such, our study contributes to the existing literature on customer concentration and sheds light on the economic consequences of customer concentration in the context of its impact on stock price crash risk.

Furthermore, recent literature investigates various determinants of future stock price crash risk. This literature has documented that opaque financial reporting, executive equity incentives, CEO overconfidence, corporate tax avoidance, and transient institutional ownership are positively associated with firm-specific stock price crash risk, while it decreases with accounting conservatism, corporate social responsibility, industry-specialist auditors, the adoption of International Financial Reporting Standards, religiosity, and financial statement comparability (Hutton et al., 2009; Kim et al., 2011a, 2011b; An & Zhang, 2013; Kim, Li, & Li, 2014; Callen & Fang, 2015; Robin & Zhang, 2015; DeFond, Hung, Li, & Li, 2015; Kim, Li, Li, & Yu, 2016a; Kim, Wang, & Zhang, 2016b; Kim & Zhang, 2016). We contribute to the stock price crash risk literature by showing that customer concentration is an important determinant of the future stock price crash risk of supplier firms, along the supplier chain.

The paper proceeds as follows. In Sections 2 and 3, we review the relevant literature and develop our research hypotheses. In Section 4, we discuss our research design. In Section 5, we present the empirical results. Section 6 concludes the paper.

## 2. Literature review

### 2.1. Customer concentration

Relying heavily on a few major customers for a large portion of sales can be risky for suppliers. First, if any major customer becomes financially distressed or switches to a different supplier, or develops product internally, a supplier firm can lose significant sales from this major customer, which could in turn lead to a huge decline in the suppliers' cash flow (Dhaliwal et al., 2016, 2019; Huang et al., 2016). Consistent with this reasoning, prior studies (Hertzel, Li, Officer, & Rodgers, 2008; Kolay, Lemmon, & Tashjian, 2016) document that suppliers experience negative abnormal returns in response to any declaration of bankruptcy from their major customers. Second, suppliers can face the risk of losing future cash flows from any inability to collect outstanding receivables if major customers become bankrupt (Dhaliwal et al., 2016). This notion is consistent with the evidence that a supplier offering its major customers more trade credit experiences greater abnormal negative abnormal returns in response to any declaration of bankruptcy announcement from a major customer (Jorion & Zhang, 2009; Kolay et al., 2016). Third, a supplier firm can be exposed to higher potential business risk even when the supplier firm is able to keep financially stable major customers. Major customers, who are naturally powerful players within the supply chain, are likely to demand stricter requirements and unique designs for their products, when compared to diversified non-

[3] The sample of Patatoukas (2012) excludes firms with negative operating earnings whereas that of Irvine et al. (2016) includes such firms.

major customers. In this environment, a supplier firm may have to invest heavily in relationship-specific assets whose alternative use is limited (Dhaliwal et al., 2019). Furthermore, major customers with strong bargaining power are likely to extract significant price concessions and trade credit extensions from suppliers, which may lead to a significant decrease in supplier profitability (Schumacher, 1991; Snyder, 1996). Fourth, a supplier with high customer concentration can face higher demand uncertainty and higher fixed-cost structure (lower cost elasticity) which leads to an increase in operating risk (Irvine et al., 2016). This occurs because they could face greater exposure to demand shocks generated by their major customers and make greater fixed investments which are specific to their relationships with major customers. Such customer-specific investments allow suppliers to meet the increased demand of major customers, but, when demand falls, these customer-specific fixed investments are more difficult to eliminate or transfer to other customers, relative to other general investments (Irvine et al., 2016). Consistent with high risks associated with customer concentration, Dhaliwal et al. (2016) document a positive relationship between customer concentration and suppliers' cost of equity and cost of public debt. Campello and Gao (2017) find a positive association between customer concentration and suppliers' loans with higher interest rate spreads, more restrictive covenants, and shorter maturities.

A contrasting view is that a supplier firm with a greater reliance on a few major customers may have lower business risks; such firms can reduce operating expenses like inventory and administrative costs and enhance asset utilization due to customer-supplier relationships along the supply chain (Kalwani & Narayandas, 1995; Patatoukas, 2012). A concentrated customer base is likely to foster information sharing along the supply chain, and facilitate product coordination and collaborative marketing and advertising efforts, which engenders efficiencies in production and marketing costs, as well as inventory and working capital management (Patatoukas, 2012). Moreover, major customers can provide an additional layer of monitoring to reduce overall risks underlying the supply-chain relationships (Fee, Hadlock, & Thomas, 2006). In a similar vein, Patatoukas (2012) documents a positive association between customer concentration and accounting-based performance measures (return on asset and return on equity).

### 2.2. Stock price crash risk

Stock price crash risk is the likelihood that an abrupt, large-scale decline in a firm's stock price occurs, and is typically captured by the third moment of firm-specific return distribution or the conditional skewness of distribution of firm-specific returns after netting out the common (market-wide and/or industry-wide) factors. The prominent predictor of stock price crash risk is the managerial tendency to withhold bad news from outside investors. Prior studies (Hutton et al., 2009; Jin & Myers, 2006; Kim et al., 2011a, 2011b; Kothari, Shu, & Wysocki, 2009) argue that managers withhold bad news from external stakeholders because of their concerns over job security, and incentive compensation based on short-term performance, aggressive tax planning and strategies. However, bad news cannot be concealed forever. When hidden bad news has accumulated over a tipping point beyond which the cost of hiding bad news is greater than the associated benefit, all bad news that has built up inside the supplier firm comes out all at once, leading to a large-scale, unanticipated decline in stock price or simply a stock price crash. Related to this notion, managers, on average, tend to delay the disclosure of bad news to investors (Kothari et al., 2009). A survey result consistently shows that managers with bad news tend to delay its disclosure more than those with good news (Graham, Harvey, & Rajgopal, 2005).

## 3. Hypotheses development

Based on prior studies and our reasoning, we develop our first hypothesis related to the association between corporate customer concentration and future stock price crash risk. On the one hand, corporate customer concentration can increase stock price crash risk for the following reasons. First, a supplier firm with major corporate customers faces significant business and cash flow risks, if any major corporate customer becomes distressed or switches to a different supplier. In addition, a supplier firm with a concentrated corporate customer base tends to engage in relationship-specific investments because managers of the supplier firm have foresight regarding the potential benefits of making such investments (i.e., increase in demand from a strong customer) (Banker et al., 2014; Kang et al., 2009).[4] Examples of such investments include investing in specialized inventory management systems, hiring quality customer representatives, enhancing delivery capabilities, and purchasing machines to tailor to specific customer needs (Chang, Hall, & Paz, 2017). Such investments are generally risky, particularly when the supplier-customer relationship becomes unstable or adversely affected by uncontrollable environmental factors, because only a long-term explicit contract with customers allows a supplier firm to definitively recoup investment costs.[5]

Furthermore, relationship-specific investments are more likely to be negative net present value (NPV) projects that entail short-term losses and are more difficult to abandon, particularly when suppliers rely on a few major corporate customers and the suppliers' investment decisions depend heavily on their customers' preferences, but not their own. In this vein, suppliers with a concentrated corporate customer base are likely to engage in bad news hoarding, raising significant business and cash flow risks and, as a result, they experience a higher level of stock price crash risk.

Second, a supplier firm may have incentive to manage its earnings upward to maintain its relationship with major corporate customers because those customers will evaluate financial performance of the supplier firm and assess the risk of maintaining current business relations with the supplier (Raman & Shahrur, 2008).[6] Furthermore, a supplier firm which relies on a few major corporate customers may have incentive to manage perceptions of these major customers, particularly the negative perceptions on the supplier firm's ability to generate future cash flows and manage business risk in relation to major customers (Dhaliwal et al., 2016; Huang et al., 2016). Hence, managers of such supplier firms are more likely to engage in bad news hoarding through earnings management to manage customers' expectations, which in turn increases its future stock price crash risk.[7]

Moreover, when a supplier firm with a concentrated corporate customer base is engaged in relationship-specific investments, the value of risky relationship-specific investments can be maintained and enhanced by the supplier and its customers' favourable prospects. If either party goes default or reneges on the terms of trade, the supplier and its customers likely end or modify their long-term business relations and suffer losses from relation-specific investments (Dou, Hope, & Thomas, 2013; Raman & Shahrur, 2008). Hence, a supplier firm may have

[4] Irvine et al. (2016) find that firms with higher concentration of large corporate customers make greater relationship-specific investments early in the customer relationship and are initially more likely to incur losses. However, as the relationship matures, the initial negative effect reverses, leading to lower cost of credit, increasing technology sharing, and improvement in firm profitability.

[5] Terms of future transactions are very uncertain and outcomes from relation-specific investments are not perfectly describable, ex-ante, and are not contractible (Tirole, 1999).

[6] Bowen, DuCharme, and Shores (1995) suggest that the firm selects income-increasing accounting discretions when implicit claims between the firm and its stakeholders such as customers and employees are higher.

[7] Suppliers relying on major customers are more likely to be involved in aggressive tax avoidance activities (Huang et al., 2016; Cen, Mayhew, Zhang, & Zuo, 2017) that can be used to manage firms' earnings upward (Dhaliwal, Gleason, & Mills, 2004; Cook, Huston, & Omer, 2008).

incentive to inflate its earnings to enhance the perception of major corporate customers. Consistent with this argument, Raman and Shahrur (2008) find that relation-specific investments between a supplier firm and its customers are positively associated with the magnitude of discretionary accruals and the likelihood of reporting a large earnings increase. This could in turn contribute to building up stock price crash risk through bad news hoarding in the future.

On the other hand, corporate customer concentration can decrease stock price crash risk. First, a supplier's reliance on a few major customers may lead to a more efficient supplier-customer relationship, a decrease in administrative and inventory costs, enhanced working capital, and thus an improvement in the supplier's financial performance (Irvine et al., 2016; Patatoukas, 2012), which in turn reduces business and cash flow risks faced by the supplier firm. In this vein, a supplier firm with a concentrated corporate customer base, which has lower business and cash flow risks, is likely to experience a lower level of bad news hoarding and stock price crash risk. Second, a supplier with a concentrated corporate customer base should be more responsive to the demand for higher quality accounting practices in general[8] and conservative accounting practices in particular, because such practices facilitate external monitoring of the supplier by major customers who hold a bargaining advantage (Hui et al., 2012).[9] Given that accounting conservatism reduces stock price crash risk (Kim & Zhang, 2016) as it deters bad news hoarding and overinvestment, a supplier firm with a concentrated corporate customer base is likely to experience a lower level of stock price crash risk.

Therefore, how corporate customer concentration affects future stock price crash risk is ultimately an empirical question. To provide large-sample, systematic evidence on this unresolved issue, we propose and test our first hypothesis, stated in a non-directional form:

**H1**. *Corporate customer concentration of a supplier firm is not associated with its future stock price crash risk.*

This far, we have discussed the impact of corporate customer concentration on suppliers. In sharp contrast to corporate customer concentration, relying on a large portion of sales on government customers may entail reduced business risk to suppliers for the following reasons (Dhaliwal et al., 2016; Huang et al., 2016). First, government customers (government or government-related customers) are unlikely to declare bankruptcy and provide stable cash flows to suppliers. Second, government customers typically make long-term contracts with their suppliers, which reduces the possibility that government customers will switch to a different supplier. Third, a substantial fraction of governmental contracts use cost-plus pricing. This contractual arrangement typically increases operational risk to government customers, but not to their suppliers. Consistent with the reasoning that government customers entail a lower risk, Dhaliwal et al. (2016) document a negative association between government customer concentration and the suppliers' cost of equity. In a similar vein, Huang et al. (2016) show that suppliers with government customers engage in lower levels of tax avoidance. Furthermore, Cohen and Li (2013) find that a supplier with a high government customer concentration hold less cash because of higher stability of operating cash flows. Thus, the reduced cash flows and business risk related to government customers are likely to discourage suppliers' bad news hoarding behaviours. We therefore expect to observe a negative relation between the government customer concentration of a supplier and its stock price crash risk. To provide large-sample, systematic evidence on this unexplored issue, we propose and test our second hypothesis, stated in the alternative form:

**H2.** *The government customer concentration of a supplier firm is negatively associated with its future stock price crash risk.*

## 4. Research design

### 4.1. Sample selection

We first obtain the supplier-customer relationship data from the Compustat segment customer files to construct our customer concentration measures. Next, we collect the stock return data from the Center for Research in Security Prices (CRSP) files to estimate future stock price crash risk. We collect the financial statement data from the Compustat annual files and the stock trading data from CRSP daily stock files. We use both accounting and market measures to measure the financial performance and stock price movement of our sample firms. We exclude firms in the utility (SIC4900-4999) and financial service (SIC6000-6900) industries from our sample.[10] After we merge our customer concentration measures with the stock price crash risk measures using unique firm-year pairs, we remove the firm-year observations with missing values for both the accounting and market performance measures. Our sample period begins in 1977 when the first disclosure on major customers was required by supplier firms in their annual reports and ends in 2016 which is most recent year for which the supplier-customer relationship data was available from the Compustat segment customer files at the time of our data collection. Our selection procedures leave us with a final sample of 131,505 firm-year observations, consisting of 13,612 unique supplier firms over the period of 40 years between 1977 and 2016. The actual numbers of observations used in our study can be slightly different, depending on regression model specifications, since the data availability varies across the analyses.

### 4.2. Customer concentration risk measures

Since the Financial Accounting Standard Board (FASB) issued the Statement of Financial Accounting Standards No. 14 (SFAS 14) in 1976, a firm with major customers which individually represent more than 10% of the firm's revenues has been required to disclose the identity and type[11] of each major customer along with the amount of revenue derived from each of them in its annual reports. From the Compustat segment customer files, we first identified supplier firms which disclosed the sales to their major customers. As some supplier firms voluntarily disclose information regarding customers which account for less than 10% of sales in their annual reports, we omitted those customers to mitigate the possible self-selection bias associated with voluntary disclosure in the calculations of our customer concentration measures, following Dhaliwal et al. (2016).

Based on prior studies (Dhaliwal et al., 2016; Patatoukas, 2012), we construct three measures of corporate (government) customer concentration, Major Corp (Major Gov't), Corp (Gov't) Customer HHI, and Major Corp (Gov't) Customer Sales to capture the degree to which a supplier firm depends on its major corporate customers in terms of

[8] As concentrated major customers monitor the suppliers' operations and induce them to improve efficiency and productivity, suppliers likely manage earnings less and this could lead to a lower level of crash risk. This is similar to the positive impact of governance function such as product competition on earnings management and disclosure quality (Li, 2010; Laksmana & Yang, 2014).

[9] When customers are concentrated and have a bargaining advantage, suppliers recognize bad news more quickly than good news. In other words, a concentrated customer base could lead to more conservative accounting practices (Hui et al., 2012), which in turn decrease future stock price crash risk.

[10] These firms are probably less prone to the agency conflict because of the intense regulatory pressure. Thus, the adverse effect of customer concentration found in industrial firms are not likely to be significant in these industries.

[11] Compustat segment customer files provide seven customer types including corporate and government customer (state, local, domestic, and foreign government).

**Table 1**
Summary statistics. Table below presents the summary statistics for customer concentration measures and stock price crash risk measures along with control variables. Definitions of all variables are presented in Appendix A. All variables are winsorized at both upper and lower 1% to prevent extreme outliers from distorting our estimations.

| Variable | mean | p50 | sd | p25 | p75 |
|---|---|---|---|---|---|
| *Panel A. Customer Concentration* | | | | | |
| Major Corp Customer | 0.374 | 0.000 | 0.484 | 0.000 | 1.000 |
| Corp Customer HHI | 0.052 | 0.000 | 0.132 | 0.000 | 0.040 |
| Major Corp Customer Sales | 0.150 | 0.000 | 0.247 | 0.000 | 0.231 |
| Major Gov't Customer | 0.077 | 0.000 | 0.266 | 0.000 | 0.000 |
| Gov't Customer HHI | 0.016 | 0.000 | 0.088 | 0.000 | 0.000 |
| Major Gov't Customer Sales | 0.032 | 0.000 | 0.133 | 0.000 | 0.000 |
| *Panel B. Stock Crash Risk* | | | | | |
| CRASH | 0.185 | 0.000 | 0.388 | 0.000 | 0.000 |
| DUVOL | −0.038 | −0.052 | 0.392 | −0.287 | 0.192 |
| NCSKEW | −0.040 | −0.082 | 0.869 | −0.491 | 0.341 |
| *Panel C. Control Variables* | | | | | |
| $MB_{t-1}$ | 2.584 | 1.718 | 3.733 | 1.003 | 3.022 |
| $LEV_{t-1}$ | 0.228 | 0.200 | 0.201 | 0.049 | 0.347 |
| $SIZE_{t-1}$ | 5.273 | 5.141 | 2.272 | 3.538 | 6.848 |
| $ROA_{t-1}$ | −0.020 | 0.037 | 0.216 | −0.026 | 0.078 |
| $ABACC_{t-1}$ | 0.444 | 0.220 | 0.656 | 0.100 | 0.492 |
| $DTURNOVER_{t-1}$ | 0.002 | 0.000 | 0.082 | −0.018 | 0.019 |
| $SIGMA_{t-1}$ | 0.064 | 0.054 | 0.037 | 0.038 | 0.080 |
| $RET_{t-1}$ | −0.003 | −0.001 | 0.003 | −0.003 | −0.001 |
| $NCSKEW_{t-1}$ | −0.048 | −0.089 | 0.790 | −0.491 | 0.326 |

annual revenues as described in detail in Appendix B. In Panel A of Table 1, the mean value of Major Corp (Gov't) Customer indicates that about 37.4% (7.7%) of our firm-year observations have at least one major corporate (government) customer who accounts for more than 10% of a supplier firm's annual revenues. Additionally, Major Corp (Gov't) Customer Sales indicates that on average, major corporate (government) customers contribute to 15.0% (3.2%) of annual revenue of the supplier firms, over our sample period from 1977 to 2016.

### 4.3. Stock crash risk measures

Following prior research (e.g., Hutton et al., 2009; Kim et al., 2011a, 2011b), we construct three firm-specific measures of stock price crash risk, *CRASH, NCSKEW,* and *DUVOL* as detailed in Appendix C. As presented in Panel B of Table 1, the mean value of CRASH is 18.5%, which is comparable to that reported in Kim et al. (16.1%, 2011a and 17.2%, 2011b). The mean *DUVOL* and *NCSKEW* is −0.038 an −0.040, respectively, which are slightly lower than those reported in Kim et al. (2011a, 2011b), perhaps due to the expanded sample period of our study.

### 4.4. Control variables

Control variables which affect future stock price crash risk have been identified in prior studies (Chen et al., 2001; Hutton et al., 2009; Kim & Zhang, 2016; Kim et al., 2011a, 2011b, 2014). Following Chen et al. (2001), who document that crash risk is higher for stocks with high investor opinion heterogeneity, high stock volatility, high positive past returns, and high past return skewness, we control for stock-specific characteristics by including (1) de-trended stock trading volume (*DTURNOVER*) for the intensity of differences of opinions among investors, (2) firm-specific weekly return volatility (*SIGMA*) as volatile stocks are more likely to be crash-prone, (3) firm-specific average weekly returns (*RET*) for potential bubble build-up as indicated by high past returns and subsequent large price drop, and (4) prior-year negative skewness (*NCSKEW*) for the potential serial correlation of the third moment of stock returns. We also control the market-to-book ratio (MB) because growth stocks with high past returns could experience high crash risk and firm size (SIZE), calculated by the natural log of total assets, to control for the positive firm size effect documented in prior studies (Chen et al., 2001; Hutton et al., 2009). Opacity, measured by the prior three years' moving sum of the absolute value of performance-matched discretionary accruals (Kothari, Leone, & Wasley, 2005), is a proxy for the managers' general propensity to hide bad information and firms with opaque financial statements experience greater crash risk (Hutton et al., 2009). In addition, financial leverage (*LEV*), calculated by total assets minus stockholders' equity, divided by the total assets, is likely to increase the monitoring intensity of managers' bad news hoarding by creditors and a negative association between leverage and crash risk is expected. High profitability (ROA) is likely to reduce the managers' bad news hoarding and a negative association between ROA and crash risk is expected.

### 4.5. Empirical Models

To test our hypotheses, we estimate the following baseline model in Eq. (1). Each of the three crash risk measures, i.e., the dependent variable, is measured in year *t*, while all the independent variables are measured in year *t-1*. In all models, we also include year-fixed-effects to address the possible effect of economic fluctuation during our sample period and industry-fixed-effects (defined by two-digit SIC code) to address the fact that some industries are more likely to suffer from stock price crash risk, than other industries.

$$Crash\ \ Measure_{,t} = \alpha + \beta_1 * Customer\ \ Concentration\ \ Measur\ e_t - 1 + \beta_2 * Controls_{j,t-1} + Industry - fixed\ effect + Year - fixed\ \ effects + \varepsilon \quad (1)$$

Depending on the type of the crash risk measure used as the dependent variable, we employ logit/OLS regression models. Specifically, we employ logit regression models when crash risk is measured by *CRASH*. We use an OLS regression if *CRASH RISK* is measured by either *NCSKEW* or *DUVOL*. We use three customer concentration measures, whether government or corporate (*Major Customer, Customer HHI* and *Major Customer Sales*) as CUSTOMER CONCENTRATION MEASURES in Eq. (1). We include firm or stock-specific control variables in Eq. (1). Appendix A provides the detailed definitions of all the variables used in our regression analyses.

## 5. Empirical results

### 5.1. Tests of hypothesis H1: The effect of corporate customer concentration on crash risk

Table 2 presents the empirical results for the tests of our first hypothesis (H1) relating to the association between the corporate customer concentration of a supplier firm and its future stock price crash risk. Throughout the paper, all reported t-values in parentheses are based on the robust standard errors corrected for double (firm and year) clustering. In Table 2, Panel A presents the results of the logit regression for Eq. (1) with CRASH in year *t* as the dependent variable. As shown in column 1, the coefficient of Major Corp Customer is positive and highly significant, suggesting that a supplier firm with at least one major corporate customer is more likely to experience a stock crash at least once a year, when compared to its competitors who do not have major corporate customers. This finding implies that a supplier firm with at least one major corporate customer is more likely to withhold bad news related to cash flows and business risks and earnings management, thereby increasing its stock price crash risk.

Models 2 and 3, where we replace *Major Corp Customer* with *Corp Customer HHI* and *Major Corp Customer Sales*, also exhibit positive and significant coefficients on *Corp Customer HHI* and *Major Corp Customer Sales*, corroborating the detrimental impact that corporate customer concentration has on crash risk (as reported in column 1). In order to

**Table 2**
The effect of corporate customer concentration on stock price crash risk. Panel A presents the results of Logit regressions which examine the impact of corporate customer concentration measures on stock price crash risk. CRASH has a value of 1 if a firm experiences one or more crash events in a year, and 0 otherwise. Panel B presents results of OLS regressions to examine the impact of corporate customer concentration measures on stock crash risk. In Models 1 to 3, DUVOL measures the down-to-up volatility of crash likelihood. In Models 4 to 6, NCSKEW represents the negative conditional return skewness. All regressions include industry and year dummies. T-statistics clustered at both firm and year are in parentheses. * p < 0.1, ** p < 0.05, *** p < 0.01.

| Panel A<br>LOGIT Models | (1)<br>CRASH | (2) | (3) |
|---|---|---|---|
| Major Corp Customer$_{t-1}$ | 0.083*** | | |
| | (4.659) | | |
| Corp Customer HHI$_{t-1}$ | | 0.274*** | |
| | | (4.483) | |
| Major Corp Customer Sales$_{t-1}$ | | | 0.180*** |
| | | | (5.243) |
| MB$_{t-1}$ | 0.009*** | 0.009*** | 0.009*** |
| | (4.354) | (4.334) | (4.322) |
| LEV $_{t-1}$ | 0.029 | 0.033 | 0.033 |
| | (0.667) | (0.758) | (0.771) |
| SIZE $_{t-1}$ | 0.013** | 0.011** | 0.013** |
| | (2.439) | (2.204) | (2.428) |
| ROA $_{t-1}$ | 0.158*** | 0.172*** | 0.167*** |
| | (3.511) | (3.832) | (3.715) |
| ABACC $_{t-1}$ | 0.019 | 0.016 | 0.016 |
| | (1.388) | (1.192) | (1.210) |
| DTURNOVER $_{t-1}$ | 0.438*** | 0.438*** | 0.439*** |
| | (4.484) | (4.483) | (4.501) |
| SIGMA $_{t-1}$ | 3.287*** | 3.313*** | 3.203*** |
| | (3.696) | (3.726) | (3.599) |
| RET $_{t-1}$ | 47.434*** | 47.808*** | 46.936*** |
| | (5.414) | (5.457) | (5.357) |
| NCSKEW $_{t-1}$ | 0.079*** | 0.079*** | 0.079*** |
| | (7.531) | (7.554) | (7.545) |
| Constant | −3.211*** | −3.206*** | −3.211*** |
| | (−15.561) | (−15.555) | (−15.577) |
| Observations | 109,055 | 109,055 | 109,055 |
| Year FE | YES | YES | YES |
| Industry FE | YES | YES | YES |
| Pseudo $R^2$ | 0.0281 | 0.0280 | 0.0281 |

| Panel B<br>OLS Models | (1)<br>DUVOL | (2) | (3) | (4)<br>NCSKEW | (5) | (6) |
|---|---|---|---|---|---|---|
| Major Corp Customer$_{t-1}$ | 0.015*** | | | 0.030*** | | |
| | (5.651) | | | (5.059) | | |
| Corp Customer HHI$_{t-1}$ | | 0.052*** | | | 0.115*** | |
| | | (4.967) | | | (4.625) | |
| Major Corp Customer Sales$_{t-1}$ | | | 0.036*** | | | 0.076*** |
| | | | (6.512) | | | (5.859) |
| MB$_{t-1}$ | 0.002*** | 0.002*** | 0.002*** | 0.004*** | 0.004*** | 0.004*** |
| | (6.211) | (6.191) | (6.164) | (5.408) | (5.375) | (5.359) |
| LEV $_{t-1}$ | 0.016** | 0.017** | 0.017*** | 0.032** | 0.034** | 0.034** |
| | (2.438) | (2.546) | (2.588) | (2.159) | (2.281) | (2.304) |
| SIZE $_{t-1}$ | 0.029*** | 0.029*** | 0.029*** | 0.062*** | 0.062*** | 0.062*** |
| | (35.368) | (35.180) | (35.463) | (33.690) | (33.572) | (33.807) |
| ROA $_{t-1}$ | 0.021*** | 0.024*** | 0.023*** | 0.035** | 0.041** | 0.039** |
| | (2.997) | (3.360) | (3.251) | (2.147) | (2.498) | (2.379) |
| ABACC $_{t-1}$ | 0.006** | 0.005** | 0.005** | 0.011** | 0.010* | 0.010* |
| | (2.493) | (2.290) | (2.267) | (2.100) | (1.901) | (1.898) |
| DTURNOVER $_{t-1}$ | 0.098*** | 0.098*** | 0.098*** | 0.207*** | 0.207*** | 0.207*** |
| | (6.435) | (6.432) | (6.456) | (5.892) | (5.891) | (5.912) |
| SIGMA $_{t-1}$ | 1.307*** | 1.311*** | 1.288*** | 3.568*** | 3.568*** | 3.526*** |
| | (9.320) | (9.352) | (9.182) | (11.392) | (11.398) | (11.255) |
| RET $_{t-1}$ | 11.763*** | 11.825*** | 11.650*** | 31.833*** | 31.899*** | 31.573*** |
| | (8.854) | (8.908) | (8.772) | (10.768) | (10.801) | (10.686) |
| NCSKEW $_{t-1}$ | 0.019*** | 0.019*** | 0.019*** | 0.041*** | 0.041*** | 0.041*** |
| | (11.007) | (11.036) | (11.025) | (10.316) | (10.345) | (10.333) |
| Constant | −0.408*** | −0.407*** | −0.408*** | −0.842*** | −0.841*** | −0.843*** |
| | (−16.686) | (−16.683) | (−16.720) | (−16.522) | (−16.525) | (−16.558) |
| Observations | 108,879 | 108,879 | 108,879 | 108,917 | 108,917 | 108,917 |
| Year FE | YES | YES | YES | YES | YES | YES |
| Industry FE | YES | YES | YES | YES | YES | YES |
| Adjusted $R^2$ | 0.0592 | 0.0591 | 0.0593 | 0.0482 | 0.0482 | 0.0484 |

assess the economic magnitude of this effect, we estimate the marginal effect of each of corporate customer concentration measure on the expected probability of a future stock crash to occur. While holding all other control variables at their means, we find that the marginal effects of *Major Corp Customer, Corp Customer HHI,* and *Major Corp Customer Sales* are 1.2%, 4%, and 2.6% respectively, implying that the effect of corporate customer concentration on the occurrence of future stock crash is not only statistically significant, but also economically meaningful, given that the unconditional mean of CRASH is 18.5% in our sample.

Panel B of Table 2 reports results of the OLS regressions where DUVOL and NCSKEW, are used as the dependent variables[12] in columns 1 to 3 and columns 4 to 6, respectively, and the same corporate customer concentration measures are used as our test variables of interest. As shown in columns 1 to 3 (4 to 6) in Panel B of Table 2, both the down-to-up volatility (DUVOL) and the negative conditional return skewness (NCSKEW) are positively and significantly related to each of the three corporate concentration measures at less than the 1% level, which is consistent with the results shown in Panel A of Table 2. When it comes to economic significance, having at least one major corporate customer results in significant increases in both continuous measures of future stock crash risk. Specifically, DUVOL and NCSKEW rise by 0.015 and 0.03 which represent increases in DUVOL and NCSKEW of 3.8% (0.015/0.392) and 3.5% (0.03/0.869) of their respective standard deviations when a supplier firm has at least one major corporate customer. These findings lend additional support to our first hypothesis H1, indicating that firms with highly concentrated corporate customer bases are exposed to a high degree of future stock price crash risk.

With respect to the control variables in Table 2, we find that their coefficients are generally in line with the findings of prior studies. Specifically, *DTURNOVER* (Chen et al., 2001) and *NCSKEW* (Kim et al., 2011a, 2011b) are significantly positive in all models in Table 2. *SIZE* is a firm-specific control variable whose positive coefficient is highly significant in all cases, which is consistent with prior findings in the current body of literature (e.g., Hutton et al., 2009). Unlike prior studies (e.g., Kim, Wang, & et al., 2016b; Callen & Fang, 2015), we find that coefficients of *ROA* are positive in most cases. This finding suggests that investors might factor potential reversal in firm profitability over time into their assessment of the current stock price.

Overall, our findings in Table 2 provide fairly consistent evidence that the corporate customer concentration of a supplier firm is positively associated with its future stock price crash risk after controlling for firm and stock-specific characteristics along with year and industry fixed effects. The association between the two is highly robust, not only to the use of different crash risk measures, but also to the use of different corporate customer concentration measures.

### 5.2. Tests of hypothesis H2: The effect of government customer concentration on crash risk

To test our second hypothesis (H2) that the government customer concentration of a supplier firm is negatively associated with its future stock price risk, we employ the same empirical model presented in Eq. (1) but replace our measures of corporate customer concentration with those of government customer concentration as our test variables of interest. We include each of the corporate customer concentration measures corresponding to each of the government customer concentration measures as an additional control variable to isolate[13] the effect of government customer concentration from the observed effect in Table 2 of corporate customer concentration on future stock crash risk in Eq. (2) below.

$$Crash\ \ Measure_{j,t} = \alpha + \beta_1 * Corporate\ \ Customer\ \ Concentration\ \ Measure_{j,t-1} + \beta_2 * Government\ \ Customer\ \ Concentration\ \ Measure_{j,t-1} + \beta_3 * Control\ s_{j,t-1} + Industry - fixed\ \ effect + Year - fixed\ effects + \varepsilon \quad (2)$$

As shown in Table 3, we find that all three government customer concentration measures in year *t-1* are negatively related to stock price crash risk in year *t*, while each corresponding corporate customer concentration measure remains significantly positive in all models. Specifically, as presented in column 1 of Panel A, the coefficient on *Major Gov't Customer* is negative and highly significant (t = -2.274), suggesting that a supplier firm with at least one major government customer is less likely to experience a future stock price crash. This finding is consistent with our expectation that a supplier firm who is dependent on safer government customers as a significant source of its annual revenue is likely to be less risky and more stable in terms of future cash flow and business prospects. Therefore, this kind of supplier firm is less likely to withhold bad news related to cash flow and business risks, when it is associated with major government customers' business. When we replace *Major Gov't Customer* with *Gov't Customer HHI* and *Major Gov't Customer Sales*, we continue to find negative and significant coefficients on *Gov't Customer HHI* and *Major Gov't Customer Sales*, suggesting that a supplier firm with high government customer concentrations experiences a lower level of future crash risk. In terms of economic magnitude, we estimate the marginal effect of each of government customer concentration measures on the expected probability of future stock crash. While holding all other control variables at their means, we find that the marginal effects of *Major Gov't Customer, Gov't Customer HHI,* and *Major Gov't Customer Sales on having at least one crash week occur in a subsequent year (CRASH)* are −1.1%, −2.9%, and −2.0% respectively, implying that the beneficial effect of having governments as customers on the occurrence of future stock price crash is not only statistically significant, but also economically meaningful. We further expand our analyses using the OLS regressions with the two continuous measures for future stock price crash risk, *DUVOL* and *NCSKEW* as the dependent variables. Similar to Panel A in Table 3 (where CRASH is used as the dependent variable), the results in Panel B show the significantly negative coefficients of all three government customer concentration measures, irrespective of whether crash risk is measured by *DUVOL* or *NCSKEW*. These results are also consistent with the findings in Dhaliwal et al. (2016) who document a negative association between government customer concentration of a supplier firm and its cost of equity capital. Overall, our findings in Panels A and B of Table 3 lend strong support to our second hypothesis H2 – that a supplier firm with higher government customer concentration is less exposed to business risk, and thus, has less incentives to hoard bad news and this therefore lowers their future stock price crash risks.

Additionally, results in Table 3 imply that the effect of major corporate customer concentration on future stock crash risk of a supplier firm might vary, depending on whether the supplier firm also has major government customer(s) at the same time[14]. To further test this implication empirically, we create one interaction term between each corporate customer concentration measure and each government customer concentration measure and include it in Eq (2). The coefficient of the interaction term is designed to capture the varying effect of major corporate customers of a supplier firm on its stock crash risk, depending on the concurrent existence and degree of major government customers

[12] As a robustness test, we use the number of crash weeks in a year as an additional measure for stock price crash risk and find that our results remain robust (not tabulated).

[13] It is possible that having higher government customer concentration can simply capture the fact that the firm has less corporate customer concentration.

[14] We appreciate the reviewer for suggesting that we investigate the potential interaction effect between corporate and government customer concentrations.

**Table 3**
The effect of government customer concentration on stock price crash risk. Panel A presents the results of Logit regressions which examine the impact of government customer concentration measures on stock price crash risk. CRASH has a value of 1 if a firm experiences one or more crash events in a year, and 0 otherwise. Panel B presents results of OLS regressions to examine the impact of government customer concentration measures on stock price crash risk. In Models 1 to 3, DUVOL measures the down-to-up volatility of crash likelihood. In Models 4 to 6, NCSKEW represents the negative conditional return skewness. All regressions include industry and year dummies. T-statistics clustered at both firm and year are in parentheses. * p < 0.1, ** p < 0.05, *** p < 0.01.

| Panel A<br>LOGIT Models | (1)<br>CRASH | (2) | (3) |
|---|---|---|---|
| Major Gov't Customer$_{t-1}$ | −0.073** | | |
| | (−2.274) | | |
| Major Corp Customer$_{t-1}$ | 0.084*** | | |
| | (4.685) | | |
| Gov't Customer HHI$_{t-1}$ | | −0.198** | |
| | | (−2.076) | |
| Corp Customer HHI$_{t-1}$ | | 0.281*** | |
| | | (4.584) | |
| Major Gov't Customer Sales$_{t-1}$ | | | −0.136** |
| | | | (−2.135) |
| Major Corp Customer Sales$_{t-1}$ | | | 0.181*** |
| | | | (5.263) |
| Observations | 109,055 | 109,055 | 109,055 |
| Control Variables | YES | YES | YES |
| Year FE | YES | YES | YES |
| Industry FE | YES | YES | YES |
| Pseudo $R^2$ | 0.0281 | 0.0281 | 0.0282 |

| Panel B<br>OLS Models | (1)<br>DUVOL | (2) | (3) | (4)<br>NCSKEW | (5) | (6) |
|---|---|---|---|---|---|---|
| Major Gov't Customer$_{t-1}$ | −0.016*** | | | −0.029*** | | |
| | (−3.395) | | | (−2.756) | | |
| Major Corp Customer$_{t-1}$ | 0.015*** | | | 0.031*** | | |
| | (5.716) | | | (5.111) | | |
| Gov't Customer HHI$_{t-1}$ | | −0.041*** | | | −0.080** | |
| | | (−2.930) | | | (−2.524) | |
| Corp Customer HHI$_{t-1}$ | | 0.054*** | | | 0.119*** | |
| | | (5.117) | | | (4.744) | |
| Major Gov't Customer Sales$_{t-1}$ | | | −0.028*** | | | −0.053** |
| | | | (−2.970) | | | (−2.507) |
| Major Corp Customer Sales$_{t-1}$ | | | 0.037*** | | | 0.076*** |
| | | | (6.568) | | | (5.904) |
| Observations | 108,879 | 108,879 | 108,879 | 108,917 | 108,917 | 108,917 |
| Control Variables | YES | YES | YES | YES | YES | YES |
| Year FE | YES | YES | YES | YES | YES | YES |
| Industry FE | YES | YES | YES | YES | YES | YES |
| Adjusted $R^2$ | 0.0592 | 0.0592 | 0.0594 | 0.0483 | 0.0483 | 0.0484 |

of the supplier firm. As shown in Appendix D, we find that corporate customer concentration measures continue to be significantly positive in all models while the interaction terms are negative and significant in five out of nine models, suggesting that, in general, the detrimental effect of corporate customer concentration on stock crash risk can be attenuated by the beneficial effect of government customer concentration.

### 5.3. *Cross-sectional tests*[15]*: Do switching costs and information asymmetries matter?*

#### 5.3.1. *The moderating effect of customer switching costs*

Prior studies (Dhaliwal et al., 2016; Huang et al., 2016) suggest that a supplier firm with a concentrated corporate customer base is exposed to a high risk of losing its major corporate customers, as major customers are able to switch their suppliers from one to another without incurring high switching costs. As a result, major corporate customers with low switching costs are likely to have higher bargaining power against suppliers and, consequently, can act more effectively as external monitors and encourage suppliers to disclose sufficient information, good or bad, in a timely manner to improve their monitoring efficacy. As long as the stock market is information-efficient, then timely-disclosed bad news would be incorporated promptly into the stock price, thereby contributing to a decreased future stock price crash risk for the supplier firm. On the other hand, low switching costs for major customers also enables them to easily switch their suppliers and thereby increase the business and cash flow risks of the suppliers and their subsequent bad news hoarding and crash risks. Thus, the predicted direction of the moderating effect of low switching costs for major customers on the association between customer concentration and stock crash risk is not ex-ante clear. Following Dhaliwal et al. (2016), we use a supplier firm's market share in its 2-digit SIC industry to measure the switching cost faced by major corporate customers. Specifically, a supplier firm's market share is defined as the ratio of its sales to the total 2-digit SIC industry sales in year *t-1*.[16] Then, we construct an

[15] We further examined the moderating role of relationship-specific investment by a supplier firm. As shown in Appendix H, we did not find clear evidence that the effect of corporate customer concentration of a supplier firm on its future stock crash risk can vary, depending on the intensity of relationship-specific investments that the supplier firm makes for its major corporate customers.

[16] As suggested in Dhaliwal et al. (2016), we also use the 3-digit SIC industry data to measure an alternative switching cost and find that our results are robust to this alternative measure of switching cost (not tabulated).

**Table 4**
The moderating effect of customer switching costs. The table below reports the interaction effect of customer switching costs and customer concentration measures on stock price crash risk. Panel A presents results of Logit regressions where CRASH is the dependent variable while Panel B presents results of OLS regressions where DUVOL and NCSKEW are the dependent variables. Low Switching Cost has a value of 1 if customer switching costs are below the sample median market share of all firms in the same 2-digit SIC industry in a given year, and 0 otherwise. Customer switching costs are measured using supplier firms' market share based on 2-digit SIC industries. All regressions include industry and year dummies. Z- or T-statistics clustered at both firm and year are in parentheses, depending on the type of regression models. * p < 0.1, ** p < 0.05, *** p < 0.01.

| Panel A<br>LOGIT Models | (1)<br>CRASH | (3) | (5) |
|---|---|---|---|
| Major Corp Customer$_{t-1}$ | 0.116*** | | |
| | (5.262) | | |
| Major Corp Customer$_{t-1}$ X Low Switching Cost | −0.085** | | |
| | (−2.486) | | |
| Corp Customer HHI$_{t-1}$ | | 0.384*** | |
| | | (4.047) | |
| Corp Customer HHI$_{t-1}$ X Low Switching Cost | | −0.174 | |
| | | (−1.453) | |
| Major Corp Customer Sales$_{t-1}$ | | | 0.245*** |
| | | | (5.359) |
| Major Corp Customer Sales$_{t-1}$ X Low Switching Cost | | | −0.133** |
| | | | (−2.060) |
| Low Switching Costs | 0.021 | −0.010 | 0.005 |
| | (0.743) | (−0.398) | (0.200) |
| Observations | 109,055 | 109,055 | 109,055 |
| Control Variables | YES | YES | YES |
| Year FE | YES | YES | YES |
| Industry FE | YES | YES | YES |
| Pseudo $R^2$ | 0.0281 | 0.0281 | 0.0282 |

| Panel B<br>OLS Models | (1)<br>DUVOL | (2) | (3) | (4)<br>NCSKEW | (5) | (6) |
|---|---|---|---|---|---|---|
| Major Corp Customer$_{t-1}$ | 0.023*** | | | 0.048*** | | |
| | (7.117) | | | (6.620) | | |
| Major Corp Customer$_{t-1}$ X Low Switching Cost | −0.020*** | | | −0.043*** | | |
| | (−3.795) | | | (−3.639) | | |
| Corp Customer HHI$_{t-1}$ | | 0.092*** | | | 0.191*** | |
| | | (5.561) | | | (5.047) | |
| Corp Customer HHI$_{t-1}$ X Low Switching Cost | | −0.061*** | | | −0.115** | |
| | | (−2.924) | | | (−2.359) | |
| Major Corp Customer Sales$_{t-1}$ | | | 0.055*** | | | 0.114*** |
| | | | (7.491) | | | (6.787) |
| Major Corp Customer Sales$_{t-1}$ X Low Switching Cost | | | −0.037*** | | | −0.075*** |
| | | | (−3.520) | | | (−3.070) |
| Low Switching Costs | 0.004 | −0.002 | 0.001 | 0.005 | −0.008 | −0.001 |
| | (0.947) | (−0.435) | (0.366) | (0.573) | (−0.966) | (−0.128) |
| Observations | 108,879 | 108,879 | 108,879 | 108,917 | 108,917 | 108,917 |
| Control Variables | YES | YES | YES | YES | YES | YES |
| Year FE | YES | YES | YES | YES | YES | YES |
| Industry FE | YES | YES | YES | YES | YES | YES |
| Adjusted $R^2$ | 0.0593 | 0.0592 | 0.0594 | 0.0484 | 0.0483 | 0.0485 |

indicator variable *Low Switching Cost* that equals 1 if a supplier firm's market share is *below* the sample median market share of all firms, excluding the supplier firm in the same two-digit SIC industry in a given year, and 0 otherwise. To examine the moderating effect of switching costs, we interact Low Switching Cost with the three corporate customer concentration measures. Specifically, we augment Eq. (1) by adding *Low Switching Cost* and its interaction with each of the three corporate customer concentration measure as follows, in Equation (3).

$$Crash\ \ Measure_{j,t} = \alpha + \beta_1 * Customer\ \ Concentration\ \ Measure_{j,t-1} + \beta_2 * Low\ \ Switching\ \ Cost_{j,t-1} + \beta_3 * Customer\ \ Concentration\ \ Measure_{j,t-1} * Low\ \ Switching\ \ Cost_{j,t-1} + \beta_4 * Controls_{j,t-1} + Industry - fixed\ \ effect + Year - fixed\ \ effects + \varepsilon. \quad (3)$$

Panel A of Table 4 presents the results of the logit regression models with CRASH as the dependent variable. As shown in columns 1 to 3 of, Panel A, Table 4, the coefficients on the interaction of *Low Switching Cost* with all three corporate customer concentration measures are negative and mostly significant, suggesting that the likelihood of a supplier firm with a concentrated corporate customer base having a higher future stock price crash risk is attenuated when its major corporate customers are able to easily switch to another supplier without incurring much cost. Furthermore, as shown in Panel B where *DUVOL* and *NCSKEW* are used as the dependent variables in columns 1 to 3 and columns 4 to 6, respectively, we again find significant and negative coefficients on the interaction terms across all six columns. Interestingly, the coefficients on the interaction terms are comparable to those of our customer concentration measures in magnitude, but with opposite signs in almost all models in Panels A and B of Table 4, suggesting that the detrimental effect of customer concentration can be almost offset by the opposite effect of the interaction term. This implies that the incentives for a supplier firm's managers to hide bad information

relating to major corporate customers can be discouraged by major corporate customers who command greater bargaining power over the supplier firm. Overall, the findings in Table 4 lend further support to the evidence that a supplier firm which has a concentrated corporate customer base becomes more information-transparent when major corporate customers have a better ability to switch supplier firms at a relatively low cost. In other words, major corporate customers which have a relatively low switching cost can act as an additional external monitoring mechanism which prevents the supplier firm with a concentrated corporate customer base from hoarding unfavourable information or delaying its disclosure (or selectively disclosing it).

#### 5.3.2. The moderating effect of information asymmetry

Information intermediation by financial analysts adds value to the market by collecting, interpreting, and discovering information (Daniel, Lee, & Naveen, 2014). In order to facilitate the information flow between firms and investors, financial analysts keep track of information disclosure by managers and periodically publish research reports which aim to help reduce the information asymmetry between inside managers and outside investors. If a firm's manager tends to disclose good news early, but withholds bad news until the last moment, then financial analysts are faced with greater information asymmetry with respect to bad news, resulting in less accuracy and wider variation in their EPS forecasts. In such cases, the firm's stock is likely to be overvalued due to bad news hoarding for a substantial period until it crashes down to its equilibrium price at the very moment when managers are forced to abandon all concealed bad news, which is sometimes accumulated over multiple periods. Therefore, we predict that the detrimental effect of corporate customer concentration on future stock price crash risk is more pronounced when a supplier firm is subject to greater information asymmetry. To test this notion, we use, as a proxy for information asymmetry, analyst forecast dispersion in a year[17], which is measured by dividing the standard deviation of annual EPS forecasts[18] by the absolute value of the mean annual EPS forecasts among financial analysts following a supplier firm. Analysts' EPS forecasts are widely distributed to outside stakeholders, particularly when relevant information is not available in a timely manner for them to analyse it properly. Specifically, following Dhaliwal et al. (2016), we construct an indicator variable *Low Dispersion* which equals 1 if dispersion of analyst annual EPS forecasts is below the median of all firms in the 2-digit SIC industry in a given year, and 0 otherwise. By construction, low dispersion represents low information asymmetry.

To examine our prediction of the moderating effect of information asymmetry, we interact *Low Dispersion* with the three corporate customer concentration measures, and re-estimate Eq. (1) by including *Low Dispersion* and its interaction with each of the three corporate customer concentration measures. Panel A of Table 5 presents the results of the logit regression with CRASH as the dependent variable. As shown in Panel A, the coefficients on the interaction terms of *Low Dispersion* with all three corporate customer concentration measures are significantly negative in columns 1 and 3, while it is negative, but insignificant in column 2. In Panel B of Table 5 where *DUVOL* or *NCSKEW* is used as the dependent variable, we estimate OLS regressions and find similar results: the three corporate customer concentration measures are significantly positive, and the coefficient on the interactions of *Low Dispersion* with the three customer concentration measures are negative and significant across all cases. Overall, the findings in Table 5 suggest that the detrimental impact of corporate customer concentration on future stock price crash risk is more pronounced for firms with high information asymmetry.

### 5.4. Robustness tests

Even though we include as many firm and stock-specific control variables as possible in the empirical models which have been identified in prior studies to affect future stock crash risk, we are not free from endogeneity concerns including omitting variables that might affect at the same time both a supplier firm's customer concentration and its stock price crash risk. For example, if our customer concentration measures are capturing nonlinear effects of our control variables on the stock price crash risk, then the linear combination of control variables in our analyses might not be sufficient to adequately take it into account the differences between a supplier firm with a concentrated corporate customer base and those without one.

### 5.5. Propensity score matching

To estimate causal treatment effects, we conduct propensity score matching (PSM) as follows. For each supplier firm which reports at least one major corporate customer (treatment group), we look for a matched control firm that does not have any major corporate customers in a given year (control group) without replacement. To this end, we first predict the likelihood that a firm has at least one major corporate customer in the first-stage logit model where Major Corp Customer, an indicator variable, is regressed on all nine firm and stock-specific control variables in Eq. (1) along with year and industry fixed effects. Using the predicted likelihood or the propensity score estimated in the first-stage logit model, we one-to-one match each firm in our treatment group with a firm in our control group that has the closest propensity score within 0.1% width without replacement. Our propensity score matching leaves us with the PSM subsample of 71,692 firm-years which consist of 35,846 supplier firm-year pairs with at least one major corporate customer and 35,846 pairs without any major corporate customer.

Following Dhaliwal et al. (2016), we conduct three matching diagnostic tests to evaluate whether our matching procedure is successful as follows. First, we estimate the logic models where the treatment group is regressed on the same control variables used for propensity score matching and compare the results for the full sample and those for the propensity score matching (PSM) subsample and find that the statistically insignificant coefficients of all control variables and the dramatic drop of pseudo-$R^2$ to below 0.1% in the PSM subsample indicate that none of the control variables predict which supplier firms are more likely to be in the treatment group. Second, we compare the summary statistics of the propensity scores including the mean and the standard deviation between the treatment and the control group and find no notable differences in all summary statistics between the two groups, suggesting that the firms in those two groups are identical in terms of propensity scores. Lastly, the mean difference tests report no statistical differences in means of all the control variables between the treatment and the control group, suggesting that firms in the two groups are identical at least in terms of firm and stock-specific variables used in our empirical models. When combined together, each of the three diagnostic tests (not reported here for brevity) strongly support that our matching procedures are successful, and the two groups essentially differ only in whether supplier firms have at least one major corporate customer or not. Therefore, if corporate customer concentration does not matter, the two groups of firms would not exhibit any significant difference in their future stock crash risks. Using this PSM subsample, we re-estimate the OLS/LOGIT regressions depending on the type of a dependent variable. Consistent with earlier findings, the results in Appendix E using the PSM subsample show that all three corporate

[17] I/B/E/S provides the mean and the standard deviation of annual EPS forecasts for multiple fiscal years as well as quarterly and semi-annual EPS forecasts at monthly frequency. We first calculate analyst forecast dispersion each month and then average them to get analyst forecast dispersion in a year in order to smooth out monthly fluctuation of analyst forecast dispersion within a year.

[18] As in Garfinkel (2009), we use annual EPS forecasts for the current fiscal year because these tend to be more accurate than EPS forecasts for later years.

**Table 5**
The moderating effect of information asymmetry. The table below reports the interaction effect of dispersion of analysts' EPS forecast and customer concentration measures on stock price crash risk. Panel A presents the results of Logit regressions where CRASH is the dependent variable while Panel B presents the results of OLS regressions where DUVOL and NCSKEW are the dependent variables. Low Dispersion has a value of 1 if dispersion of annual EPS forecasts of analysts following a firm is below 2-digit SIC industry median dispersion in a given year, and 0 otherwise. Following Garfinkel (2009), dispersion is measured as standard deviations of annual EPS forecasts, scaled by absolute mean forecasts of all analysts following a firm in a given year. To be included in the sample, the firm should be followed by at least two analysts in a given year. All regressions include industry and year dummies. Z- or T-statistics clustered at both firm and year are in parentheses depending on the type of regression models. * p < 0.1, ** p < 0.05, *** p < 0.01.

| Panel A<br>LOGIT Models | (1)<br>CRASH | (3) | (5) |
|---|---|---|---|
| Major Corp Customer$_{t-1}$ | 0.115*** | | |
| | (3.935) | | |
| Major Corp Customer$_{t-1}$ X Low Dispersion | −0.077** | | |
| | (−1.962) | | |
| Corp Customer HHI$_{t-1}$ | | 0.298*** | |
| | | (3.083) | |
| Corp Customer HHI$_{t-1}$ X Low Dispersion | | −0.083 | |
| | | (−0.538) | |
| Major Corp Customer Sales$_{t-1}$ | | | 0.223*** |
| | | | (4.129) |
| Major Corp Customer Sales$_{t-1}$ X Low Dispersion | | | −0.135* |
| | | | (−1.744) |
| Low Dispersion | 0.051** | 0.027 | 0.044* |
| | (2.056) | (1.269) | (1.905) |
| Observations | 71,723 | 71,723 | 71,723 |
| Control Variables | YES | YES | YES |
| Year FE | YES | YES | YES |
| Industry FE | YES | YES | YES |
| Pseudo $R^2$ | 0.0340 | 0.0340 | 0.0341 |

| Panel B<br>OLS Models | (1)<br>DUVOL | (2) | (3) | (4)<br>NCSKEW | (5) | (6) |
|---|---|---|---|---|---|---|
| Major Corp Customer$_{t-1}$ | 0.023*** | | | 0.048*** | | |
| | (5.238) | | | (4.936) | | |
| Major Corp Customer$_{t-1}$ X Low Dispersion | −0.015*** | | | −0.038*** | | |
| | (−2.622) | | | (−2.894) | | |
| Corp Customer HHI$_{t-1}$ | | 0.078*** | | | 0.172*** | |
| | | (4.453) | | | (4.108) | |
| Corp Customer HHI$_{t-1}$ X Low Dispersion | | −0.042* | | | −0.106* | |
| | | (−1.679) | | | (−1.687) | |
| Major Corp Customer Sales$_{t-1}$ | | | 0.050*** | | | 0.105*** |
| | | | (5.668) | | | (5.090) |
| Major Corp Customer Sales$_{t-1}$ X Low Dispersion | | | −0.030** | | | −0.074** |
| | | | (−2.450) | | | (−2.572) |
| Low Dispersion | 0.005 | 0.002 | 0.005 | 0.013* | 0.005 | 0.010 |
| | (1.570) | (0.727) | (1.423) | (1.668) | (0.699) | (1.441) |
| Observations | 71,694 | 71,694 | 71,694 | 71,697 | 71,697 | 71,697 |
| Control Variables | YES | YES | YES | YES | YES | YES |
| Year FE | YES | YES | YES | YES | YES | YES |
| Industry FE | YES | YES | YES | YES | YES | YES |
| Adjusted $R^2$ | 0.0528 | 0.0528 | 0.0530 | 0.0418 | 0.0418 | 0.0419 |

customer concentration measures still carry positive and significant coefficients, irrespective of which stock price crash measure is used. Thus, the results using the PSM subsample both support and enrich the conclusion that corporate customer concentration leads supplier firms to experience a higher incidence and a greater degree of future stock price crash. In summary, the results are robust to potential endogeneity and show that customer concentration is not merely associated with, but rather brings about a higher incidence and a greater degree of future stock price crash risk.

### 5.6. Fixed-effect models

It is conceivable that our results in Table 2 might be driven by certain unobservable firm characteristics that are omitted in the models. One approach that can help mitigate the effect of the omitted-variable bias is to run a firm fixed-effect model[19]. This approach controls for any unobservable characteristics that remain time-invariant over our sample period because it captures only the variation over time within firms. We, therefore, re-estimate the baseline OLS regressions in Eq (1), but replace industry fixed effects with firm fixed effects. The

[19] We perform the change analysis to examine the impact of changes in one-year period in corporate customer concentration on changes in future stock price crash risk over the same period. Given that there is not much variation in *CRASH* in one-year period, we focus on two continuous stock price crash risk measures, *DUVOL* and *NCSKEW* as dependent variables. All three corporate customer concentration measures still carry an expected positive sign, and their coefficients are all significant, irrespective of how future stock price crash risk is measured, suggesting that changes in stock price crash risk are not just associated with, but driven by changes in customer concentration. Results will be available upon request to authors.

regression results in Appendix F report that all three corporate customer concentration measures still carry an expected positive sign, and their coefficients are mostly significant (seven out of nine models), irrespective of how stock price crash risk is measured, after controlling for firm-fixed effects. Therefore, our results are unlikely driven by unobservable and time-invariant firm characteristics that might be omitted.

### 5.7. Instrumental-variable analysis (IV)

In order to mitigate endogeneity, we execute an instrumental-variable (IV) analysis[20]. This approach alleviates the endogeneity biases that can be attributed to reverse causality and unobserved heterogeneity. Following prior studies (Hilary & Hui, 2009; Kumar et al., 2011) who use an earlier year's value of their focal explanatory variable as an IV, we employ customer concentration in the earliest year for each firm as our IV[21], assuming that it affects customer concentration in any subsequent years of the same firm (relevance condition), but could influence the firm's stock crash risk in subsequent years only through its impact on subsequent years' customer concentration (exclusion condition). For each firm, we identify the earliest year when the firm appears in the sample. Then, we use customer concentration in the earliest year as our IV. One possible criticism against using the earliest value as an instrument is that customer concentration might be sticky, changing only slowly over time. If this is the case, then the value in the earliest year may not be so different from the value in any of the subsequent years. To alleviate this concern, we take the following steps. We calculate the standard deviation of corporate customer concentration for each firm over time. A higher standard deviation indicates that corporate customer concentration in the firm changes more quickly, while a low standard deviation implies that it is sticky over time. In the second-stage IV regression, we include only those firms whose corporate customer concentration changes more quickly than the median. This way, we concentrate only on those firms whose corporate customer concentration changes more quickly, mitigating the concern for stickiness. Results (Model (1) in Panel A of Appendix G) from the first-stage IV regression where Corp Customer HHI is the dependent variable shows that the coefficient of Corp Customer HHI (Earliest year) is positive and highly significant. In the second-stage IV regression where DUVOL and NCSKEW are used respectively as the dependent variable, the coefficient of Corp Customer HHI predicted/instrumented from the first stage IV regression is significantly positive as shown in Models (2) and (3) in Panel A of Appendix G. Therefore, our IV analysis confirms that greater corporate customer concentration leads to higher stock price crash risk. As a robustness check, when we perform a similar analysis using Major Corp Customer Sales as our measure of corporate customer concentration, we find very similar results as reported in Models (4), (5), and (6). The high F-statistics refute the null hypothesis that our instrument is weak, while the statistically significant Durbin p-values also suggest that there might be an endogenous relationship between corporate customer concentration of a frim and its stock crash risk, supporting the validity of employing the IV estimation. Furthermore, to ensure that the results are robust, we adopt an alternate instrumental variable. We posit that customers and suppliers located closer to each other are more likely to have a relationship as they are expected to be aware of and interact with each other. In other words, locations matter. If this is the case, then firms in the same area should attract similar customers and exhibit similar customer concentration (relevance condition). As a result, we utilize the average corporate customer concentration in the same U.S. county[22], excluding firm i, as our alternative IV. Because we exploit the corporate customer concentration of the surrounding firms without including firm *i*, our instrument is likely exogenous (exclusion condition). Each firm has control over its own customer concentration, but it does not have any control over the customer concentration of another firm in the same area. As shown in Panel B of Appendix G, results using this alternative instrumental variable indicate that corporate customer concentration exacerbates the stock price crash risk, corroborating the earlier documented results. Because the IV analysis is more robust to endogeneity, our results are more likely to show a causal effect of corporate customer concentration on the stock price crash risk.

### 5.8. Mediating models

So far, we have documented robust evidence that the corporate customer concentration of a firm leads directly to a significantly higher likelihood that the firm will experience a stock price crash in the future. However, Huang et al. (2016) argue that supplier firms with a concentrated corporate customer base are more likely to engage in aggressive tax avoidance activities to manipulate major customers' perception of its financial health while Kim, Li, and Zhang (2011b) document that corporate tax avoidance activities lead to firm-specific stock price crash risks. Therefore, some might argue that our documented effect of corporate customer concentration on stock crash risk is merely picking up an indirect effect of corporate customer concentration via corporate tax avoidance on stock crash risk. To mitigate this concern, we further extend our research question to understand whether corporate tax avoidance acts as a channel/mechanism through which corporate customer concentration affects stock crash risk, and if so, to what degree. In other words, we test whether and to what degree a firm's tax avoidance can mediate the observed positive effect of a firm's corporate customer concentration on its stock crash risk. To separate the indirect effect of corporate customer concentration via the firm's tax avoidance behaviour from its direct effect on stock crash risk, we establish the mediation models where we employ OLS regressions where two of the stock crash risk measures (DUVOL and NCSKEW) are regressed on each corporate customer concentration measure while corporate tax avoidance behaviour[23] is included as a mediating variable. In our setting, the direct effect captures an effect of corporate customer concentration (causal variable, CV) on stock crash risk (outcome variable) that is not mediated by corporate tax avoidance (mediating variable, MV) and the indirect effect is an effect of corporate customer concentration on stock crash risk that travels through corporate tax avoidance. The total effect represents the sum of the direct and the indirect effects. As shown in Table 6, the $CurrentETR_{t-1}$ representing the degree of corporate tax avoidance[24] exhibits a significantly negative coefficient, suggesting that aggressive tax avoidance

[20] IV analysis is widely employed in finance and economics to eliminate endogeneity concerns, which include, but are not limited to (1) omitted variable bias, (2) reverse causality bias, and (3) errors-in-variable bias.

[21] We also exploit the fact that customer concentration in the earliest year for each firm could not have resulted from the stock price crash risk in the subsequent years.

[22] We use ZIP Code Tabulation Area (ZCTA) Relationship File available from US Census Bureau which provides mapping table between zip code of a firm's headquarter and FIPS code of a US county where the firm operates.

[23] To capture the degree of corporate tax avoidance, we use current effective tax rate (*CurrentETR*). Based on Huang et al. (2016), *CurrentETR* is defined as total income tax expense (Compustat: TXT), net of deferred income tax expense (Compustat: TXDI), divided by pretax net income (Compustat: PI) less special items (Compustat: SPI). By construction, *CurrentETR* reflects how effectively firms reduce their tax burden in a current period not only through permanent book-tax difference, but also by temporarily deferring it to future period, which will result in an increase in tax obligation later. Therefore, a lower *CurrentETR* implies higher level of tax avoidance.

[24] Even after we use two other traditional measures of corporate tax avoidance, book effective tax rate (Chen, Chen, Cheng, & Shevlin, 2010) and cash effective tax rate (Hanlon & Shevlin, 2003) in place of current effective tax rate, the results remain the same. Results will be available upon request to authors.

**Table 6**
The mediating effects of corporate tax avoidance. The table below reports the results of the mediation models where we examine how and whether the effect of corporate customer concentration on stock price crash risk is mediated through corporate tax avoidance. Specifically, we employ OLS regressions where the stock price crash risk is regressed on each corporate customer concentration measure while the current effective tax rate (currentETR), representing the degree of corporate tax avoidance by a firm is used as a mediating variable. All models include year dummies and industry (2 digit SIC) dummies. For definitions of other variables, please refer to Appendix A. The t-statistics are reported in parentheses. *, **, *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

| VARIABLES | (1) DUVOL | (2) DUVOL | (3) DUVOL | (4) NCSKEW | (5) NCSKEW | (6) NCSKEW |
|---|---|---|---|---|---|---|
| Major Corp Customer$_{t-1}$ | 0.013*** | | | 0.025*** | | |
| | (4.508) | | | (3.863) | | |
| Corp Customer HHI$_{t-1}$ | | 0.047*** | | | 0.100*** | |
| | | (4.462) | | | (4.253) | |
| Major Corp Customer Sales$_{t-1}$ | | | 0.032*** | | | 0.064*** |
| | | | (5.491) | | | (4.968) |
| CurrentETR $_{t-1}$ | −0.041*** | −0.041*** | −0.041*** | −0.090*** | −0.089*** | −0.089*** |
| | (−6.571) | (−6.553) | (−6.502) | (−6.393) | (−6.364) | (−6.325) |
| Observations | 89,154 | 89,154 | 89,154 | 89,185 | 89,185 | 89,185 |
| Control Variables | YES | YES | YES | YES | YES | YES |
| Year FE | YES | YES | YES | YES | YES | YES |
| Industry FE | YES | YES | YES | YES | YES | YES |
| R-squared | 0.061 | 0.061 | 0.061 | 0.050 | 0.050 | 0.050 |
| Adjusted $R^2$ | 0.0602 | 0.0602 | 0.0603 | 0.0493 | 0.0493 | 0.0494 |
| Sobel Z | 4.939 | 5.244 | 5.424 | 4.863 | 5.144 | 5.317 |
| Mediating Variable (MV) | CurrentETR $_{t-1}$ | CurrentETR $_{t-1}$ | CurrentETR $_{t-1}$ | CurrentETR $_{t-1}$ | CurrentETR $_{t-1}$ | CurrentETR $_{t-1}$ |
| Causal Variable (CV) | Major Corp Customer$_{t-1}$ | Corp Customer HHI$_{t-1}$ | Major Corp Customer Sales$_{t-1}$ | Major Corp Customer$_{t-1}$ | Corp Customer HHI$_{t-1}$ | Major Corp Customer Sales$_{t-1}$ |
| *Indirect effect of CV through MV(A)* | 0.0005*** | 0.0020*** | 0.0012*** | 0.0010*** | 0.0044*** | 0.0027*** |
| *Direct effect of CV(B)* | 0.0132*** | 0.0469*** | 0.0318*** | 0.0252*** | 0.1001*** | 0.0645*** |
| *Total effect(C = A + B)* | 0.0136*** | 0.0489*** | 0.0331*** | 0.0263*** | 0.1045*** | 0.0671*** |
| Ratio of Indirect effect to Total effect (A/C) | 0.0353 | 0.0413 | 0.0375 | 0.0399 | 0.0419 | 0.0402 |
| Ratio of Indirect effect to Direct effect (A/B) | 0.0366 | 0.0430 | 0.0390 | 0.0415 | 0.0438 | 0.0419 |
| Ratio of Total effect to Direct effect(C/B) | 1.0366 | 1.0430 | 1.0390 | 1.0415 | 1.0438 | 1.0419 |

leads to higher stock crash risk, which is consistent with findings in Kim et al. (2011b). Furthermore, we continue to observe the same positive coefficient of each of the corporate customer concentration measures, capturing its direct effect on stock crash risk. We also identify that its indirect effect through corporate tax avoidance is significantly positive, suggesting that corporate customer concentration, through its effect on corporate tax avoidance, indirectly affects stock crash risk positively. As shown in Table 6, however, the proportion of the indirect effect of corporate customer concentration on stock crash risk that is mediated through corporate tax avoidance out of the total effect is < 5% in all models, suggesting that corporate customer concentration acts as a direct determinant of stock crash risk, rather than a mere reflection of its indirect effect via corporate tax avoidance.

## 6. Conclusion

Our study examines how corporate and government customer concentration affects stock price crash risk. We also study the cross-sectional variation in the association between corporate customer concentration and stock price crash risk in terms of customer switching costs and information asymmetry. Our results demonstrate that various measures of corporate customer concentration have positive associations with different measures of stock price crash risk. We also find that various measures of government customer concentration have negative associations with different measures of stock price crash risk. Our findings suggest thatsuppliers with a concentrated corporate customer base have an incentive to manage customers' expectations of their performance and their earnings and they have a tendency to engage in risky relationship-specific investments that entail high business risk with their customers. As a result, managers of such supplier firms tend to engage in bad news hoarding behaviour. This hoarding behaviour leads to a positive association between corporate customer concentration and stock price crash risk. However, a supplier firm with a concentrated government customer base is less likely to hoard bad news related to business and cash flow risks because government customers provide stable cash flow and less bankruptcy risk and this tendency leads us to observe a negative association between government customer concentration and stock price crash risk. Further analyses show that our results are robust to PSM analyses, firm fixed-effect analyses and IV analyses. In the cross-sectional analyses, we find that the likelihood of a supplier firm with a concentrated corporate customer base having a higher stock price crash risk is attenuated by lower switching costs and is more pronounced when the degree of information asymmetry is high.

While our study provides evidence that customer concentration tends to increase a supplier firm's stock price crash risk, such evidence should not be interpreted in such a way that a supplier firm should avoid customer concentration. On the contrary, customer concentration can be beneficial for the supplier firm because of efficient product, inventory and supply chain management.

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## Appendix A. Variable definitions

| Variable | Definition |
|---|---|
| Major Corp (Gov't) Customer | 1 if a supplier firm reports at least one corporate(government) customer who accounts for more than 10% of its annual revenues, and 0 otherwise. |
| Corp (Gov't) Customer HHI | Sales-based Herfindahl-Hirschman Index as sum of the squares of the ratios of supplier firm revenue to major corporate (government) customer to its total revenue |
| Major Corp (Gov't) Customer Sales | Fraction of a supplier firm's annual revenues that are accounted for by all major corporate(government) customers |
| CRASH | 1 if we observe at least one crash week during a year, and 0 otherwise. |
| DUVOL | Log Ratio of the standard deviation of firm-specific abnormal weekly returns of the down weeks to that of the up weeks |
| NCSKEW | Negative of the ratio of the third moment to standard deviation cubed of firm-specific abnormal weekly returns |
| MB | Market to Book ratio. Market capitalization of shareholders' equity (PRCC_F*CSHO in Compustat) divided by the book value of shareholders' equity (CEQ in Compustat). |
| LEV | Leverage ratio, calculated as the total assets (AT in Compustat) minus the stockholders' equity (CEQ), divided by the total assets (AT in Compustat). |
| SIZE | Natural log of total assets (AT in Compustat). |
| ROA | Return On Assets. Net income (NI in Compustat) divided by total assets (AT in Compustat). |
| ABACC | Prior three years' moving sum of the absolute value of performance matched discretionary accruals estimated based on Kothari et al. (2005). |
| DTURNOVER | Average monthly share turnover for the last fiscal year minus the average monthly share turnover for the year before the last fiscal year. Monthly share turnover is defined as the total monthly trading volume (VOL in CRSP) divided by the total monthly number of shares outstanding (SHROUT in Compustat). |
| SIGMA | Standard deviation of firm-specific abnormal weekly returns |
| RET | Fiscal year's average firm-specific abnormal weekly returns |

### *Appendix B. Customer concentration risk measures*

We first create an indicator variable, *Major Corporate Customer,* which equals 1 if a supplier firm reports at least one corporate customer who accounts for more than 10% of its annual revenues, and 0 otherwise.

As our second customer concentration measure, we borrow the concept of the Herfindahl-Hirschman Index (HHI), and compute Customer HHI for major corporate customers, denoted by *Corp Customer HHI*. We compute *Corp Customer HHI* for each supplier firm *i* by summing the squares of the ratios of a supplier firm *i*'s revenue to each major corporate customer *j's* total revenue in year *t*. This *Corp Customer HHI* variable is designed to take into account not only the number of major corporate customers *j* disclosed by a supplier firm *i*, but also the influence that each major corporate customer *j* exercises on the supplier firm *i*'s revenues.[25] The specific formula we use for *Corp Customer HHI* is as follows:

$$Corp\ \ Customer\ \ HHI_{i,t} = \sum_{j=1}^{J} \left( \frac{Revenues_{i,j,t}}{Revenues_{i,t}} \right)^2 \tag{4}$$

Our third measure of corporate customer concentration, denoted by *Major Corp Customer Sales*, is defined as the fraction of a supplier firm *i*'s annual revenues that are accounted for by all major corporate customers *j* in year *t* as shown in formula below.[26]

$$Major\ \ Corp\ \ Customer\ \ Sales_{i,t} = \sum_{j=1}^{J} (Revenue_{i,j,t}/Revenue_{i,t}) \tag{5}$$

To test our second hypothesis about the effect of *government*[27] customer concentration on future crash risk, we apply the same methodology that we use for *corporate* customer concentration measures. Thus we construct three measures of government customer concentration, *Major Gov't Customer, Gov't Customer HHI,* and *Major Gov't Customer Sales,* which correspond to *Major Corp Customer, Corp Customer HHI,* and *Major Corp Customer Sales*, respectively, to capture the incidence and degree to which a supplier firm depends on its major government customers in terms of annual revenues.

## Appendix C. Stock crash risk measures

To construct each firm-specific crash risk measure, we first run the following extended index model in Eq. (3) using unique firm-year-week observations and define firm-specific weekly return for firm *j* in week *t* as $W_{j,t} = \ln(1 + \varepsilon_{j,t})$ where $\varepsilon_{j,t}$ is the residual return from Eq. (6).

$$r_{j,t} = \alpha_j + \beta_{1,j} r_{m,t-2} + \beta_{2,j} r_{m,t-1} + \beta_{3,j} r_{m,t} + \beta_{4,j} r_{m,t+1} + \beta_{5,j} r_{m,t+2} + \varepsilon_{j,t} \tag{6}$$

where $r_{j,t}$ is the return on stock *j* in week *t* and $r_{m,t}$ is the market index weighted by value from CRSP in week *t*. If a firm experiences firm-specific weekly returns $W_{j,t}$ for a specific week which is equal to or lower than 3.2 standard deviations below the mean of the firm-specific weekly returns $W_{j,t}$ for the entire fiscal year, we then define that week as a crash week (Hutton et al., 2009; Kim et al., 2011a, 2011b).[28] We first create an indicator variable, *CRASH,* which equals 1 if we observe at least one crash week during the entire year and, 0 otherwise.

As our second crash risk measure, we calculate the negative conditional firm-specific return skewness (*NCSKEW*), following prior research (Chen et al., 2001; Kim et al., 2011a, 2011b). *NCSKEW* is defined as the negative of the ratio of the third moment to the standard deviation cubed of firm-specific weekly returns, $W_{j,t}$ for each firm-year pair, as shown in formula below:

[25] By construction, this measure has a maximum value of 1 if one single major corporate customer, *j* accounts for all revenues earned by the supplier firm, *i* and a minimum of zero if a supplier firm, *i* does not report any major corporate customer, *j* in its annual reports.

[26] By construction, the higher (lower) value of Major Corp Customer Sales represents more (less) customer concentration.

[27] In defining government customer concentration measures, we include not only U.S. federal government, but also state, local, and foreign governments, given that any type of government is less likely to default than corporate customers.

[28] As in prior studies, we choose 3.2 standard deviations to generate a frequency of 0.1% in the normal distribution.

$$NCSKEW_{j,t} = -[n(n-1)^{3/2}\sum W_{j,t}^3]/[(n-1)(n-2)(\sum W_{j,t}^2)^{3/2}] \tag{7}$$

As the third stock price crash risk measure, we create the down-to-up volatility (*DUVOL*) (Chen et al., 2001; Kim et al., 2011a, 2011b). We first separate all weeks within each unique firm-year pair into the down weeks which experience firm-specific weekly returns $W_{j,t}$ below its mean of firm-specific weekly returns $W_{j,t}$ for the entire year. We then calculate the standard deviation of firm-specific weekly returns $W_{j,t}$for the down weeks. We make the same separation and calculations of the standard deviation of firm-specific weekly returns, $W_{j,t}$for the up weeks. We then define the variable, *DUVOL* as the log ratio of the standard deviation of firm-specific abnormal weekly returns, $W_{j,t}$of the down weeks to that of the up weeks in the following formula.

$$DUVOL_{j,t} = \log\left\{\frac{(n_u-1)\sum_{DOWN} W_{j,t}^2}{(n_d-1)\sum_{UP} W_{j,t}^2}\right\} \tag{8}$$

where $n_u$is the number of the up weeks and $n_d$ is the number of the down weeks for the entire year of a given firm. The higher value of *DUVOL* corresponds to a more left skewed distribution, which indicates the higher incidence of crash.

## Appendix D. The interaction effect of corporate and government customer concentration on stock price crash risk

Panel A presents the results of Logit regressions which examine the interaction impact between corporate customer concentration and government customer concentration measures on stock price crash risk. Interaction Term represents the interaction between each measure of corporate and government customer concentration used in the same model. In Panel A, CRASH has a value of 1 if a firm experiences one or more crash events in a year, and 0 otherwise. In Models 1 to 3 of Panel B, DUVOL measures the down-to-up volatility of crash likelihood. In Models 4 to 6 of Panel B, NCSKEW represents the negative conditional return skewness. All regressions include industry and year dummies. T-statistics clustered at both firm and year are in parentheses. * p < 0.1, ** p < 0.05, *** p < 0.01.

| Panel A<br>LOGIT Models | (1)<br>CRASH | (2) | (3) |
|---|---|---|---|
| Interaction Term | −0.041 | −0.562 | −0.456** |
| | (−0.663) | (−1.200) | (−2.033) |
| Major Gov't Customer$_{t-1}$ | −0.055 | | |
| | (−1.308) | | |
| Major Corp Customer$_{t-1}$ | 0.087*** | | |
| | (4.681) | | |
| Gov't Customer HHI$_{t-1}$ | | −0.133 | |
| | | (−1.224) | |
| Corp Customer HHI$_{t-1}$ | | 0.297*** | |
| | | (4.756) | |
| Major Gov't Customer Sales$_{t-1}$ | | | −0.051 |
| | | | (−0.683) |
| Major Corp Customer Sales$_{t-1}$ | | | 0.199*** |
| | | | (5.607) |
| Observations | 109,055 | 109,055 | 109,055 |
| Control Variables | YES | YES | YES |
| Year FE | YES | YES | YES |
| Industry FE | YES | YES | YES |
| Pseudo $R^2$ | 0.0281 | 0.0281 | 0.0282 |

| Panel B<br>OLS Models | (1)<br>DUVOL | (2) | (3) | (4)<br>NCSKEW | (5) | (6) |
|---|---|---|---|---|---|---|
| Interaction Term | −0.029*** | −0.046 | −0.087*** | −0.067*** | −0.177 | −0.211*** |
| | (−3.284) | (−0.794) | (−2.780) | (−3.386) | (−1.454) | (−3.025) |
| Major Gov't Customer$_{t-1}$ | −0.003 | | | 0.001 | | |
| | (−0.510) | | | (0.049) | | |
| Major Corp Customer$_{t-1}$ | 0.018*** | | | 0.037*** | | |
| | (6.380) | | | (5.847) | | |
| Gov't Customer HHI$_{t-1}$ | | −0.035** | | | −0.058 | |
| | | (−2.214) | | | (−1.600) | |
| Corp Customer HHI$_{t-1}$ | | 0.056*** | | | 0.125*** | |
| | | (5.115) | | | (4.840) | |
| Major Gov't Customer Sales$_{t-1}$ | | | −0.011 | | | −0.011 |
| | | | (−0.945) | | | (−0.456) |
| Major Corp Customer Sales$_{t-1}$ | | | 0.041*** | | | 0.086*** |
| | | | (6.998) | | | (6.386) |
| Observations | 108,879 | 108,879 | 108,879 | 108,917 | 108,917 | 108,917 |
| Control Variables | YES | YES | YES | YES | YES | YES |
| Year FE | YES | YES | YES | YES | YES | YES |
| Industry FE | YES | YES | YES | YES | YES | YES |
| Adjusted $R^2$ | 0.0593 | 0.0592 | 0.0594 | 0.0484 | 0.0483 | 0.0485 |

## Appendix E. Robustness check: Propensity score matching

Panel A presents results of Logit regressions to examine the impact of corporate customer concentration on stock price crash risk. CRASH has a value of 1 if a firm experiences one or more crash events in a year, and 0 otherwise. Panel B presents results of OLS regressions to examine the impact of corporate customer concentration on stock price crash risk. In Models 1 to 3, DUVOL measures the down-to-up volatility of crash likelihood. In Models 4 to 6, NCSKEW represents the negative conditional return skewness. All regressions include industry and year dummies. T-statistics clustered at both firm and year are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

| Panel A<br>LOGIT Models | (1)<br>CRASH | (2) | (3) |
|---|---|---|---|
| Major Corp Customer$_{t-1}$ | 0.090***<br>(4.498) | | |
| Corp Customer HHI$_{t-1}$ | | 0.266***<br>(3.785) | |
| Major Corp Customer Sales$_{t-1}$ | | | 0.185***<br>(4.771) |
| Observations | 65,951 | 65,951 | 65,951 |
| Control Variables | YES | YES | YES |
| Year FE | YES | YES | YES |
| Industry FE | YES | YES | YES |
| Pseudo $R^2$ | 0.0241 | 0.0240 | 0.0241 |

| Panel B<br>OLS Models | (1)<br>DUVOL | (2) | (3) | (4)<br>NCSKEW | (5) | (6) |
|---|---|---|---|---|---|---|
| Major Corp Customer$_{t-1}$ | 0.014***<br>(4.732) | | | 0.030***<br>(4.445) | | |
| Corp Customer HHI$_{t-1}$ | | 0.053***<br>(4.357) | | | 0.113***<br>(3.976) | |
| Major Corp Customer Sales$_{t-1}$ | | | 0.036***<br>(5.610) | | | 0.075***<br>(5.162) |
| Observations | 65,854 | 65,854 | 65,854 | 65,880 | 65,880 | 65,880 |
| Control Variables | YES | YES | YES | YES | YES | YES |
| Year FE | YES | YES | YES | YES | YES | YES |
| Industry FE | YES | YES | YES | YES | YES | YES |
| Adjusted $R^2$ | 0.0546 | 0.0546 | 0.0548 | 0.0450 | 0.0450 | 0.0452 |

## Appendix F. Robustness check: Fixed-Effect Models

Panel A presents the results of fixed-effect Logit regressions which examine the impact of corporate customer concentration on stock price crash risk. CRASH has a value of 1 if a firm experiences one or more crash events in a year, and 0 otherwise. Panel B presents results of fixed-effect OLS regressions to examine the impact of corporate customer concentration on stock price crash risk. In Models 1 to 3, DUVOL measures the down-to-up volatility of crash likelihood. In Models 4 to 6, NCSKEW represents the negative conditional return skewness. All regressions include firm- and year dummies. Robust z- or t- statistics are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

| Panel A<br>Fixed Effect LOGIT Models | (1)<br>CRASH | (2) | (3) |
|---|---|---|---|
| Major Corp Customer$_{t-1}$ | 0.051*<br>(1.800) | | |
| Corp Customer HHI$_{t-1}$ | | 0.135<br>(1.451) | |
| Major Corp Customer Sales$_{t-1}$ | | | 0.097*<br>(1.664) |
| Observations | 93,292 | 93,292 | 93,292 |
| Control Variables | YES | YES | YES |
| Year FE | YES | YES | YES |
| Firm FE | YES | YES | YES |
| Pseudo $R^2$ | 0.0293 | 0.0293 | 0.0293 |

| Panel B<br>Fixed Effect OLS Models | (1)<br>DUVOL | (2) | (3) | (4)<br>NCSKEW | (5) | (6) |
|---|---|---|---|---|---|---|
| Major Corp Customer$_{t-1}$ | 0.008**<br>(2.006) | | | 0.012<br>(1.445) | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Corp Customer $HHI_{t-1}$ | | 0.044*** | | | 0.072** | |
| | | (3.282) | | | (2.391) | |
| Major Corp Customer $Sales_{t-1}$ | | | 0.030*** | | | 0.049*** |
| | | | (3.599) | | | (2.673) |
| Observations | 108,879 | 108,879 | 108,879 | 108,917 | 108,917 | 108,917 |
| Control Variables | YES | YES | YES | YES | YES | YES |
| Year FE | YES | YES | YES | YES | YES | YES |
| Firm FE | YES | YES | YES | YES | YES | YES |
| Adjusted $R^2$ | 0.111 | 0.112 | 0.112 | 0.0994 | 0.0994 | 0.0994 |

## Appendix G. Robustness check: Instrumental-variable (IV) analysis

The table below reports the results of IV regressions. In panel A, we use corporate customer concentration in its earliest year as an instrumental variable. Specifically, Model (1) reports the 1st-stage IV regression analysis where we employ corporate customer concentration in its earliest year as an independent variable of interest while positioning corporate customer concentration in its subsequent years as a dependent variable. Models (2) and (3) report the 2nd-stage IV regression analysis where we employ customer concentration (instrumented) which is predicted in the 1st-stage IV analysis in Model (1) as an independent variable of interest while positioning each stock crash measure as a dependent variable. To measure corporate customer concentration, we use Corp Customer HHI in Models (1) to (3) and Major Corp Customer Sales in Models (4) to (6). Panel B reports results of IV regressions where we use average corporate customer concentration of all firms excluding itself in the same US county as an instrumental variable. DUVOL measures the down-to-up volatility of crash likelihood and NCSKEW represents the negative conditional return skewness. For the tests, we focus on those firms where corporate customer concentration changes more quickly over time. To identify those firms, we first calculate the standard deviation of each corporate customer concentration measure for each firm over time. Then, we select the subsample of firms where the standard deviation of each corporate customer concentration measure is above its median. All regressions include industry and year dummies. t-statistics are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

| Panel A<br>IV stage<br>Dependent variable | (1)<br>1st-stage<br>Corp Customer $HHI_{t-1}$ | (2)<br>2nd-stage<br>DUVOL | (3)<br>2nd-stage<br>NCSKEW | (4)<br>1st-stage<br>Major Corp Customer $Sales_{t-1}$ | (5)<br>2nd-stage<br>DUVOL | (6)<br>2nd-stage<br>NCSKEW |
|---|---|---|---|---|---|---|
| Corp Customer $HHI_{t-1}$(Earliest year) | 0.364*** | | | | | |
| | (89.503) | | | | | |
| Major Corp Customer $Sales_{t-1}$ (Earliest year) | | | | 0.351*** | | |
| | | | | (87.972) | | |
| Corp Customer HHIt-1(Instrumented) | | 0.093*** | 0.228*** | | | |
| | | (5.300) | (5.761) | | | |
| Major Corp Customer $Sales_{t-1}$ (Instrumented) | | | | | 0.126*** | 0.358*** |
| | | | | | (4.269) | (5.371) |
| Observations | 55,517 | 55,517 | 55,541 | 55,654 | 55,654 | 55,679 |
| Control Variables | YES | YES | YES | YES | YES | YES |
| Year FE | YES | YES | YES | YES | YES | YES |
| Industry FE | YES | YES | YES | YES | YES | YES |
| R-squared | 0.205 | 0.063 | 0.052 | 0.225 | 0.062 | 0.051 |
| IV F-stat | | 7,739 | 7,748 | | 8,011 | 8,025 |
| Durbin pval | | 0.00867 | 9.83e-05 | | 0.000667 | 3.89e-05 |

| Panel B<br>IV stage<br>Dependent variable | (1)<br>1st-stage<br>Corp Customer $HHI_{t-1}$ | (2)<br>2nd-stage<br>DUVOL | (3)<br>2nd-stage<br>NCSKEW | (4)<br>1st-stage<br>Major Corp Customer $Sales_{t-1}$ | (5)<br>2nd-stage<br>DUVOL | (6)<br>2nd-stage<br>NCSKEW |
|---|---|---|---|---|---|---|
| Average Corp Customer $HHI_{t-1}$ of all firms excluding itself in its US county | 0.097*** | | | | | |
| | (11.357) | | | | | |
| Average Major Corp Customer $Sales_{t-1}$ of all firms excluding itself in its US county | | | | 0.147*** | | |
| | | | | (18.554) | | |
| Corp Customer HHIt-1(Instrumented) | | 0.723*** | 1.577** | | | |
| | | (2.584) | (2.523) | | | |
| Major Corp Customer $Sales_{t-1}$ (Instrumented) | | | | | 0.204** | 0.441** |
| | | | | | (2.182) | (2.114) |
| Observations | 85,314 | 85,314 | 85,346 | 85,314 | 85,314 | 85,346 |
| Control Variables | YES | YES | YES | YES | YES | YES |
| Year FE | YES | YES | YES | YES | YES | YES |
| Industry FE | YES | YES | YES | YES | YES | YES |
| R-squared | 0.124 | 0.018 | 0.009 | 0.185 | 0.052 | 0.041 |
| IV F-stat | | 129 | 129.2 | | 344.3 | 344.5 |
| Durbin pval | | 0.0152 | 0.0180 | | 0.0729 | 0.0788 |

## Appendix H. The moderating effect of Relationship-specific investments

The table below reports the interaction effect of relation-specific investment by a supplier firm and its corporate customer concentration measures on stock price crash risk. Panel A presents results of Logit regressions where CRASH is the dependent variable while Panel B presents results of OLS regressions where DUVOL and NCSKEW are the dependent variables. High Relation-specific Investment has a value of 1 if R&D investments by a supplier firm is above the sample median market share of all firms in the same 2-digit SIC industry in a given year, and 0 otherwise. All regressions include industry and year dummies. Z- or T-statistics clustered at both firm and year are in parentheses, depending on the type of regression models. * p < 0.1, ** p < 0.05, *** p < 0.01.

| Panel A<br>LOGIT Models | (1)<br>CRASH | (3) | (5) |
|---|---|---|---|
| Major Corp Customer$_{t-1}$ | 0.093*** | | |
| | (4.588) | | |
| Major Corp Customer$_{t-1}$ X High Relation-specific Investment | 0.041 | | |
| | (1.021) | | |
| Corp Customer HHI$_{t-1}$ | | 0.279*** | |
| | | (3.983) | |
| Corp Customer HHI$_{t-1}$ X High Relation-specific Investment | | 0.041 | |
| | | (0.304) | |
| Major Corp Customer Sales$_{t-1}$ | | | 0.187*** |
| | | | (4.785) |
| Major Corp Customer Sales$_{t-1}$ X High Relation-specific Investment | | | 0.038 |
| | | | (0.509) |
| High Relation-specific Investment | 0.068** | 0.083*** | 0.078*** |
| | (2.167) | (3.219) | (2.748) |
| Observations | 109,055 | 109,055 | 109,055 |
| Control Variables | YES | YES | YES |
| Year FE | YES | YES | YES |
| Industry FE | YES | YES | YES |
| Pseudo $R^2$ | 0.0282 | 0.0282 | 0.0282 |

| Panel B<br>OLS Models | (1)<br>DUVOL | (2) | (3) | (4)<br>NCSKEW | (5) | (6) |
|---|---|---|---|---|---|---|
| Major Corp Customer$_{t-1}$ | 0.018*** | | | 0.036*** | | |
| | (6.029) | | | (5.355) | | |
| Major Corp Customer$_{t-1}$ X High Relation-specific Investment | 0.013** | | | 0.024* | | |
| | (2.117) | | | (1.763) | | |
| Corp Customer HHI$_{t-1}$ | | 0.051*** | | | 0.112*** | |
| | | (4.215) | | | (3.914) | |
| Corp Customer HHI$_{t-1}$ X High Relation-specific Investment | | −0.004 | | | −0.010 | |
| | | (−0.153) | | | (−0.174) | |
| Major Corp Customer Sales$_{t-1}$ | | | 0.039*** | | | 0.079*** |
| | | | (6.084) | | | (5.422) |
| Major Corp Customer Sales$_{t-1}$ X High Relation-specific Investment | | | −0.009 | | | 0.017 |
| | | | (−0.760) | | | (0.581) |
| High Relation-specific Investment | 0.001 | 0.007* | 0.005 | 0.008 | 0.019** | 0.015 |
| | (0.330) | (1.960) | (1.306) | (0.770) | (2.224) | (1.642) |
| Observations | 108,879 | 108,879 | 108,879 | 108,917 | 108,917 | 108,917 |
| Control Variables | YES | YES | YES | YES | YES | YES |
| Year FE | YES | YES | YES | YES | YES | YES |
| Industry FE | YES | YES | YES | YES | YES | YES |
| Adjusted $R^2$ | 0.0592 | 0.0592 | 0.0593 | 0.0483 | 0.0483 | 0.0484 |

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